CORRFLOW: SELF-CORRECTING FLOW MATCHING

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ABSTRACT

Flow-matching models generate samples through iterative velocity predictions, but typically do not revisit these predictions using the intermediate representations that produced them. We introduce CORRFLOW, a self-correcting flow-matching framework that reuses this information through a lightweight refiner. Under rectified-flow interpolation, the error of the one-step clean prediction equals the realized velocity residual scaled by the noise level. Motivated by this identity, the refiner learns to predict residual corrections from the base model's clean prediction, conditioning, noise level and hidden states from the same forward pass. The correction is added to the base velocity before each sampling update, so subsequent predictions are made from the corrected state. This requires no additional base-model evaluations and only one lightweight refiner evaluation per step. Our theory gives conditions under which feature reuse outperforms raw-input affine predictors, refiner training outperforms continued base-model training at matched compute, and deployed correction lowers velocity error under its own sampler law. Experiments on class-conditional image generation, text-to-video generation and image-to-video generation show consistent improvements over the corresponding base models, including gains in visual quality and physical consistency. Project Page: `https://fogradio.github.io/corrflow-project`.

1 INTRODUCTION

Flow-matching models (Lipman et al., 2023; Liu et al., 2023) generate samples by learning a velocity field that transports noise toward the data distribution. Their simple regression objective has shown strong results in image (Esser et al., 2024), video (Wan et al., 2025), 3D (Xiang et al., 2025) and robotic action (Black et al., 2024). Yet generated content still exhibits structural errors, semantic mismatches and physical inconsistencies (Zheng et al., 2025; Sun et al., 2025; Motamed et al., 2026; Zhang et al., 2025a). Recent approaches improve generation quality through large-scale training (Esser et al., 2024), representation alignment (Yu et al., 2025), self-supervised feature learning (Chefer et al., 2026) and reward-based optimization (Yuan et al., 2026). They focus on improving the model's ability by learning only from the final answer, which is limited. This limitation can be attributed, in part, to the fact that the model must *predict a single velocity to determine the next sample state, without any opportunity to review and revise its own predictions.*

We therefore exploit the property that learning should be "*refine-in-loop*", in the sense that if a student makes a mistake, e.g., in an examination, they should revisit the answer they wrote, the question they were asked, and the reasoning they followed. By examining these pieces of context together, the student can trace a mistake and determine how to revise it, rather than struggling with a final answer (Madaan et al., 2023; Zou et al., 2025). Inspired by the success of *self-refinement* in large language models (LLMs) (Madaan et al., 2023; Shinn et al., 2023), we apply this principle to flow matching. A flow-based generator already computes analogous information at each sampling step: an estimate of the clean sample→answer, the generation conditions→question and intermediate hidden states→internal state. The challenge is to define a correction target that can be supervised during training and predicted from these quantities without access to a clean reference during generation.

In this paper, we introduce CORRFLOW, a self-correcting flow-matching framework that attaches a lightweight *refiner* to the base generator. Our *core insight* is that the model's one-step clean prediction provides a concrete object for the review, with a direct mathematical connection to the correction we seek (Chen et al., 2022). Under rectified-flow interpolation, the error between this prediction and the

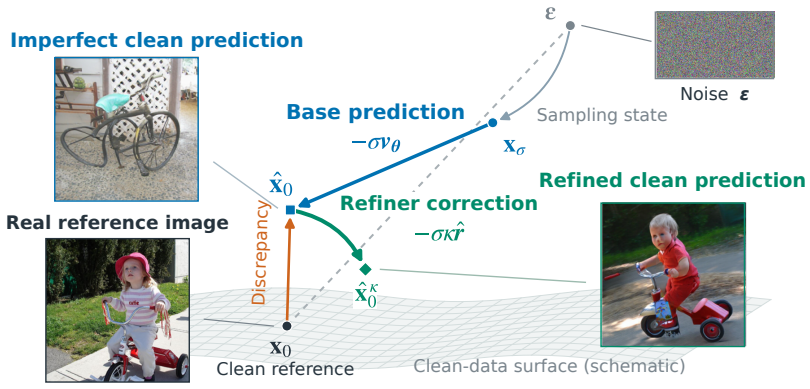

Figure 1: **CorrFlow learns from its own imperfect predictions.** The base model's clean prediction (blue) deviates from the clean reference (orange arrow). The resulting correction shifts the prediction closer to the clean reference by $-\sigma\kappa\hat{\boldsymbol{r}}$ (green arrow), where σ is the noise level and κ controls the correction strength. The initial noise is obtained by adding Gaussian noise to the clean reference.

clean sample is exactly *realized velocity residual* (eq. (4)) scaled by the noise level. This identity connects a mistake in the predicted sample to a correction in the velocity field. We first theoretically prove the relationship between the realized velocity residual and the clean prediction (section 3.1). In practice, the refiner is a transformer that predicts the velocity residual from three sources of information: *i) the current answer*, given by the one-step clean prediction, *ii) the circumstances*, given by the generation condition and noise level, and *iii) the internal state*, given by hidden states from the base model (Epstein et al., 2023). During joint training, clean samples and their sampled noise provide residual targets for the refiner, while the base model and refiner optimize separate objectives. During generation, the predicted residual adjusts the velocity used by the sampler before each state update. All refiner inputs reuse the same base-model forward pass, and each correction requires one additional refiner evaluation.

We empirically apply CorrFlow to a wide range of applications, including image generation, text-to-video generation and image-to-video generation. Figure 1 illustrates how this review shifts the model's clean prediction through a correction of its velocity on image generation. We find that training a refiner improves the base model's performance on aesthetic quality and physics consistency of generated content. The strong performance across multiple tasks and datasets shows that velocity residual correction is generic. Our contributions are summarized as follows:

- We introduce CorrFlow, a self-correcting flow-matching framework that learns velocity residuals from the base model's current prediction, and applies the resulting corrections within the sampling loop.
- We theoretically establish conditions under which reusing base-model representations outperforms raw-input affine predictors, refiner training achieves greater residual-risk reduction than continued base-model training at matched compute, and deployed correction lowers velocity error under its own sampler-induced state law.
- We evaluate CorrFlow on image and video generation benchmarks, demonstrating improvements in visual quality and physics consistency.

2 Related Works

Improving Flow-Matching Generative Models. Many approaches have been proposed to improve flow-matching generative models. For representation learning, REPA (Yu et al., 2025) aligns hidden states with pretrained visual features, while iREPA (Singh et al., 2025) emphasizes their spatial structure. VA-VAE (Yao et al., 2025) aligns VAE latents with visual features, REPA-E (Leng et al., 2025) jointly tunes the VAE and diffusion transformer through alignment and RAE (Zheng et al., 2026a) generates directly in pretrained representation spaces. Other methods reuse features available

within the generative model: SRA and LayerSync (Jiang et al., 2026; Haghighi et al., 2026) use deeper features to guide shallower representations, and SRA^2 (Wang et al., 2026a) draws on existing VAE features. Reward-based methods Flow-GRPO and DanceGRPO (Liu et al., 2025; Xue et al., 2025c) use stochastic sampling for online reinforcement learning, while AWM (Xue et al., 2025b) and DiffusionNFT (Zheng et al., 2026c) incorporate rewards into flow-matching objectives through advantage weighting and paired generations, respectively. CORRFLOW takes a complementary approach: it trains a *lightweight added-on refiner* to predict *velocity residuals* from the base model's current prediction and hidden states. These residuals directly correct the velocity within the sampling loop, without requiring external representation teachers or reward models for refiner training.

Physics-Aware Video Generation. Several recent works have explored physics-aware video generation with flow-matching models. WISA (Wang et al., 2025), PhysVid (Pathak et al., 2026) and PhyParam (Li et al., 2026) inject hierarchical, temporally local, or instance-level physical conditions. PhysGen (Liu et al., 2024) and PSIVG (Foo et al., 2026) use simulated dynamics as guidance. Representation-based methods transfer or jointly learn motion, geometry, semantics and latent dynamics, including VideoREPA (Zhang et al., 2025c), PhysAlign (Xiong et al., 2026), VideoJAM (Chefer et al., 2025), PHANTOM (Shen et al., 2026), DreamWorld (Tan et al., 2026) and MMPhysVideo (Lin et al., 2026). Other approaches use language or feedback, with PhyT2V (Xue et al., 2025a) and DiffPhy (Zhang et al., 2025b) introducing inferred physical rules, PhysMaster (Ji et al., 2025) and PhyGDPO (Cai et al., 2025) applying pairwise or groupwise preference learning and WMReward (Yuan et al., 2026) steering denoising trajectories with a latent-world-model reward at inference. In contrast, CORRFLOW learns to correct the base model's own velocity residuals from its predictions and hidden states, without explicit physical parameters or external physics evaluators.

3 CORRFLOW

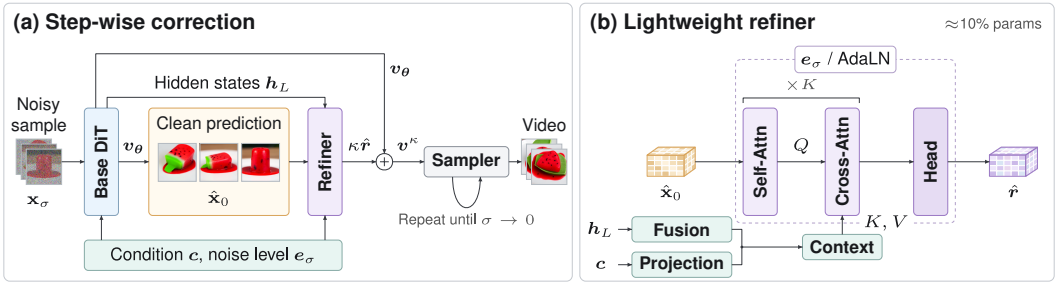

Figure 2: **CORRFLOW overview for video generation.** (a) At noise level σ, the refiner predicts a residual $\hat{\boldsymbol{r}}$ from the base DiT's clean prediction $\hat{\mathbf{x}}_0 = \mathbf{x}_\sigma - \sigma\boldsymbol{v}_{\boldsymbol{\theta}}$, selected hidden states $\boldsymbol{h}_L$, conditioning tokens $\boldsymbol{c}$ and noise-level embedding $\boldsymbol{e}_\sigma$. Sampling corrects the base velocity as $\boldsymbol{v}^\kappa = \boldsymbol{v}_{\boldsymbol{\theta}} + \kappa\hat{\boldsymbol{r}}$ with strength κ. (b) The refiner alternates self-attention over clean-prediction tokens with cross-attention to projected conditioning and fused hidden-state tokens. Noise conditioning modulates its blocks and sample-space output head, while 3D rotary embeddings encode video token coordinates. Frames show outputs and cubes denote sample tensors.

CORRFLOW augments a base flow-matching model $\boldsymbol{f}_{\text{base}}$ with a lightweight *refiner* that corrects its velocity $\boldsymbol{v}_{\boldsymbol{\theta}}$ at each sampling step. We use bold upright symbols for samples, bold italic symbols for other vectors and tensors and ordinary symbols for scalars. The refiner takes the one-step clean prediction $\hat{\mathbf{x}}_0 = \mathbf{x}_\sigma - \sigma\boldsymbol{v}_{\boldsymbol{\theta}}$, hidden states $\{\boldsymbol{h}_\ell\}_{\ell\in L}$, conditioning $\boldsymbol{c}$ and noise-level embedding $\boldsymbol{e}_\sigma$ to predict a velocity residual:

$$\hat{\boldsymbol{r}} = \boldsymbol{f}_{\text{refiner}}(\hat{\mathbf{x}}_0, \{\boldsymbol{h}_\ell\}_{\ell\in L}, \boldsymbol{c}, \boldsymbol{e}_\sigma), \qquad \boldsymbol{v}^\kappa = \boldsymbol{v}_{\boldsymbol{\theta}} + \kappa\hat{\boldsymbol{r}}, \tag{1}$$

where κ controls the correction strength.

3.1 PRELIMINARIES: FLOW MATCHING AND THE RESIDUAL IT LEAVES BEHIND

Let $\mathbf{x}_0 \in \mathbb{R}^D$ be the continuous representation of a clean data sample, with encoded conditioning $\boldsymbol{c}$, such as class or text embeddings. The representation uses the sample's native coordinates or a

learned feature space, and D denotes its dimension. Let $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, I_D)$ be independent noise in the same space. For a noise level $\sigma \in [0, 1]$ the rectified-flow interpolant and its derivative are

$$\mathbf{x}_\sigma = (1 - \sigma)\, \mathbf{x}_0 + \sigma\, \boldsymbol{\varepsilon}, \qquad \frac{\mathrm{d}\mathbf{x}_\sigma}{\mathrm{d}\sigma} = \boldsymbol{\varepsilon} - \mathbf{x}_0. \tag{2}$$

The target flow velocity is $\boldsymbol{v} = \boldsymbol{\varepsilon} - \mathbf{x}_0$, and the Bayes velocity field is $\boldsymbol{v}^\star(\mathbf{x}, \sigma, \boldsymbol{c}) = \mathbb{E}[\, \boldsymbol{\varepsilon} - \mathbf{x}_0 \mid \mathbf{x}_\sigma = \mathbf{x}, \boldsymbol{c}\,]$. The base diffusion model $\boldsymbol{f}_{\text{base}}$ is trained by

$$\mathcal{L}_{\text{base}}(\boldsymbol{\theta}) = \mathbb{E}_{\mathbf{x}_0, \boldsymbol{c}, \boldsymbol{\varepsilon}, \sigma}\, w(\sigma) \left\| \boldsymbol{v}_{\boldsymbol{\theta}}(\mathbf{x}_\sigma, \sigma, \boldsymbol{c}) - \boldsymbol{v} \right\|^2, \tag{3}$$

with $w(\sigma)$ the noise-level weight of the base recipe, where $\boldsymbol{v}_{\boldsymbol{\theta}}$ is the output of $\boldsymbol{f}_{\text{base}}$, parameterized by $\boldsymbol{\theta}$. Generation integrates the learned velocity field $\mathrm{d}\mathbf{x}/\mathrm{d}\sigma = \boldsymbol{v}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma, \boldsymbol{c})$ from $\sigma = 1$ to $\sigma = 0$ with explicit Euler steps on a shifted grid. The base model approximates the Bayes velocity field, while each sampled data–noise pair provides a realized target velocity. We call the difference between this target and the base prediction the **realized velocity residual**,

$$\boldsymbol{r} \;=\; \boldsymbol{v} - \boldsymbol{v}_{\boldsymbol{\theta}}(\mathbf{x}_\sigma, \sigma, \boldsymbol{c}) \;=\; (\boldsymbol{\varepsilon} - \mathbf{x}_0) - \boldsymbol{v}_{\boldsymbol{\theta}}(\mathbf{x}_\sigma, \sigma, \boldsymbol{c}). \tag{4}$$

This is the mistake we want the refiner $\boldsymbol{f}_{\text{refiner}}$ to predict. The base model's one-step clean prediction is available at generation time. Its discrepancy from the clean target is proportional to the realized velocity residual, as the following identity shows.

Clean-prediction identity. Taking a single Euler step from $\mathbf{x}_\sigma$ all the way to $\sigma = 0$ gives the base model's **one-step clean prediction**, its best current guess at the finished sample,

$$\hat{\mathbf{x}}_0(\mathbf{x}_\sigma, \sigma, \boldsymbol{c}) \;=\; \mathbf{x}_\sigma - \sigma\, \boldsymbol{v}_{\boldsymbol{\theta}}(\mathbf{x}_\sigma, \sigma, \boldsymbol{c}). \tag{5}$$

Substituting Equation 2 into Equation 5 and using Equation 4,

$$\hat{\mathbf{x}}_0 - \mathbf{x}_0 = \sigma\big((\boldsymbol{\varepsilon} - \mathbf{x}_0) - \boldsymbol{v}_{\boldsymbol{\theta}}\big) = \sigma\, \boldsymbol{r}, \qquad \text{equivalently} \qquad \boldsymbol{r} = \frac{\hat{\mathbf{x}}_0 - \mathbf{x}_0}{\sigma}, \quad \sigma > 0. \tag{6}$$

This identity holds for any predictor $\boldsymbol{v}_{\boldsymbol{\theta}}$ and shows that the clean-prediction error is the velocity residual $\boldsymbol{r}$ scaled by the noise level σ. It therefore motivates using $\hat{\mathbf{x}}_0$ as the refiner's primary input, expressing the prediction in the clean data space. Although $\hat{\mathbf{x}}_0$ is available at every sampling step, its error depends on the unknown clean target. The refiner thus learns to estimate the residual from this prediction $\hat{\mathbf{x}}_0$ and the accompanying features $\boldsymbol{h}_L$.

3.2 Refiner Inputs

Focusing on the current answer in isolation is insufficient, because the same prediction may be correct under one condition and a mistake under another, or an artifact that must be repaired at high noise may be a legitimate detail at low noise. Therefore, the refiner is given three views of the moment the mistake was made:

The current answer. The one-step clean prediction $\hat{\mathbf{x}}_0$ of Equation 5, tokenized according to the sample structure to form the refiner's trunk tokens. This is the base model's current estimate of the clean sample.

The circumstances. The conditioning tokens $\boldsymbol{c}$ that encode the task requirements, and the noise level σ, which identifies the current position along the denoising trajectory.

The internal state. The base model's hidden states $\{\boldsymbol{h}_\ell\}_{\ell \in L}$, where L indexes transformer blocks selected uniformly across the model's depth. These readouts capture different stages of the computation that produced $\boldsymbol{v}_{\boldsymbol{\theta}}$, and they record how the prediction was assembled rather than only what it was.

Formally, let $W = (\mathbf{x}_\sigma, \sigma, \boldsymbol{c})$ denote the base model's query state, and let $\mathcal{Z}(W) = (\hat{\mathbf{x}}_0, \{\boldsymbol{h}_\ell\}_{\ell \in L}, \sigma, \boldsymbol{c})$ be the refiner's information. The refiner is a network $\boldsymbol{f}_{\text{refiner}}$ with parameter vector $\boldsymbol{\phi}$, written as $\boldsymbol{g}_{\boldsymbol{\phi}}$, and is trained to predict the residual, $\hat{\boldsymbol{r}} = \boldsymbol{g}_{\boldsymbol{\phi}}(\mathcal{Z}(W)) \approx \boldsymbol{r}$. Every component of $\mathcal{Z}$ is from the base model's conditioning and its existing forward pass. Constructing them for the refiner requires *no additional base-model evaluation* and *no access to the clean target at generation time*.

3.3 REFINER ARCHITECTURE

CORRFLOW is designed to be a *lightweight add-on transformer* to a base flow-matching model, which estimates the realized velocity residual $\boldsymbol{r}$. It has its own parameters, totaling approximately 10% of the base model's parameter count. It carries the aforementioned information as inputs.

Token embeddings. Given the clean prediction $\hat{\mathbf{x}}_0$ (eq. (5)), the refiner $\boldsymbol{f}_{\text{refiner}}$ first tokenizes it according to the sample structure into N tokens and projects them to the refiner width d, obtaining the answer tokens $\boldsymbol{t}_{\text{ans}} \in \mathbb{R}^{N\times d}$. Our image and video implementations use patches for this tokenization. To keep the refiner and base model on a common clock, the refiner also consumes the base model's own noise-level embedding $\boldsymbol{e}_\sigma$ at the current noise level σ.

We also construct the context tokens $\boldsymbol{t}_{\text{ctx}} \in \mathbb{R}^{(N_c+N)\times d}$ from the conditioning $\boldsymbol{c}$ and the hidden states $\{\boldsymbol{h}_\ell\}_{\ell\in L}$, where N_c is the number of conditioning tokens. In particular, a linear projection maps $\boldsymbol{c}$ to the conditioning tokens $\boldsymbol{t}_{\text{c}} \in \mathbb{R}^{N_c\times d}$. At each token position, we concatenate the hidden states from the selected layers along the feature dimension and project them to the refiner width d using a multilayer perceptron with a GELU activation,

$$\boldsymbol{t}_{\text{fuse}} = \mathrm{MLP}\Big(\mathrm{concat}_{\ell\in L}\, \boldsymbol{h}_\ell\Big) \in \mathbb{R}^{N\times d}. \tag{7}$$

This fusion across layers at fixed token positions is a deliberate choice. It keeps the hidden-state tokens at N tokens instead of $|L|\cdot N$, so attention cost does not grow with the number of readouts, and it lets the refiner learn which depth to trust at each location rather than committing to a fixed average across depths. To combine the two context tokens for the refiner's cross-attention, we concatenate them along the token axis and apply layer normalization,

$$\boldsymbol{t}_{\text{ctx}} = \mathrm{LayerNorm}\Big(\big[\,\boldsymbol{t}_{\text{c}} + \boldsymbol{s}_{\text{cond}}\;,\; \boldsymbol{t}_{\text{fuse}} + \boldsymbol{s}_{\text{state}}\,\big]\Big) \in \mathbb{R}^{(N_c+N)\times d}, \tag{8}$$

where $\boldsymbol{s}_{\text{cond}}$ and $\boldsymbol{s}_{\text{state}}$ are learned segment embeddings that mark the two sources.

Alternating-Attention blocks. We build our *lightweight add-on transformer* with a stack of B Alternating Attention Blocks (AABs), making the refiner focus on the current answer while consulting the context in an alternate manner. Specifically, even-indexed blocks apply *self-attention* to the answer tokens $\boldsymbol{t}_{\text{ans}}$, using rotary position embeddings defined on the token coordinates of the sample representation. Our image and video implementations use two-dimensional spatial coordinates and three-dimensional spatiotemporal coordinates, respectively. Odd-indexed blocks apply *cross-attention* to the context tokens $\boldsymbol{t}_{\text{ctx}}$, which lets each token incorporate the task requirements and the base model's intermediate representations. Each attention layer is followed by a feed-forward network. Following the base model's adaptive-normalization design, each block uses shift, scale and gate parameters derived from the noise-level embedding $\boldsymbol{e}_\sigma$. Finally, a modulated linear head projects the updated answer tokens to residual tokens, which are reassembled into the sample's original shape.

3.4 TRAINING AND INFERENCE

Objective. We train the base model $\boldsymbol{f}_{\text{base}}$ and the refiner $\boldsymbol{f}_{\text{refiner}}$ on the same minibatch using separate objectives and a single backward pass. In particular, we define the refiner's loss by a combination of a magnitude term and a direction term,

$$\mathcal{L}_{\text{ref}}(\boldsymbol{\phi}) = \mathbb{E}\Big[\; \big\|\hat{\boldsymbol{r}} - \mathrm{sg}[\boldsymbol{r}]\big\|^2 \;+\; \lambda\big(1 - \cos\big(\hat{\boldsymbol{r}},\, \mathrm{sg}[\boldsymbol{r}]\big)\big)\Big], \tag{9}$$

where $\mathrm{sg}[\cdot]$ denotes the `stop-gradient` operator, $\hat{\boldsymbol{r}} = \boldsymbol{f}_{\text{refiner}}(\mathrm{sg}[\mathcal{Z}(W)])$ is the refiner output, with $\boldsymbol{r}$ the realized velocity residual defined in Equation 4 and λ controls the weight of the direction term. We calculate the cosine similarity by flattening the overall residual field into a single vector per sample. The two terms play complementary roles. The squared-error term fits the residual vector, coupling its magnitude and direction. When residual directions vary across samples with similar refiner inputs, averaging these targets can produce a correction with a smaller magnitude. The cosine term explicitly rewards alignment with the target direction and complements this vector regression while the correction magnitude is learned. The total training objective is

$$\mathcal{L} = \mathcal{L}_{\text{base}}(\boldsymbol{\theta}) + \lambda_{\text{ref}}\, \mathcal{L}_{\text{ref}}(\boldsymbol{\phi}), \tag{10}$$

where $\boldsymbol{\theta}$ and $\boldsymbol{\phi}$ are the base and refiner parameters, respectively. Here λ_{ref} controls the relative weight of the refiner objective.

Inference. The refiner is applied inside the ordinary sampling loop rather than as a post-process on a finished sample. At each denoising step the sampler evaluates the base model on the current sample, captures $\boldsymbol{e}_\sigma$ and $\{\boldsymbol{h}_\ell\}_{\ell\in L}$ from that same evaluation, assembles $\hat{\mathbf{x}}_0$ from Equation 5 and runs the refiner once on $\mathcal{Z}(W)$. The predicted residual is added to the base velocity with a gain κ,

$$\boldsymbol{v}^\kappa = \boldsymbol{v}_{\boldsymbol{\theta}}(\mathbf{x}_\sigma, \sigma, \boldsymbol{c}) + \kappa\, \boldsymbol{f}_{\text{refiner}}(\mathcal{Z}(W)), \tag{11}$$

and the corrected velocity is what the scheduler integrates. Because $\hat{\mathbf{x}}_0$ and $\{\boldsymbol{h}_\ell\}_{\ell\in L}$ are obtained from the state the sampler has actually reached, each correction influences subsequent refiner inputs. This adds one refiner forward pass per step without additional base-model evaluations.

4 Theoretical Analysis

We analyze three complementary questions: whether frozen base-model features can improve residual prediction beyond raw-input predictors, whether refiner training can outperform continued base-model training at matched budget and whether a trained correction improves velocity error under the state law it induces at deployment.

Shared residual target. Let $\mathcal{X}$ be a finite-dimensional Hilbert sample space, let P be the law of the query state $W = (\mathbf{x}_\sigma, \sigma, \boldsymbol{c})$ and let $\widetilde{P}$ be the joint law of $(W, \boldsymbol{r})$ at the fixed base checkpoint used below. Assume $\boldsymbol{r} \in L^2(\widetilde{P}; \mathcal{X})$ and write $\bar{\boldsymbol{r}}(W) = \mathbb{E}_{\widetilde{P}}[\boldsymbol{r} \mid W]$. For any $\boldsymbol{f} \in L^2(P; \mathcal{X})$, define

$$\mathcal{R}_P(\boldsymbol{f}) := \mathbb{E}_{\widetilde{P}}\|\boldsymbol{r} - \boldsymbol{f}(W)\|^2, \qquad \mathcal{R}_{\text{irr}} := \mathbb{E}_{\widetilde{P}}\|\boldsymbol{r} - \bar{\boldsymbol{r}}(W)\|^2.$$

Proposition 1 (Same-target risk decomposition)**.** *Every such predictor satisfies*

$$\mathcal{R}_P(\boldsymbol{f}) = \mathcal{R}_{\text{irr}} + \mathbb{E}_P\|\boldsymbol{f}(W) - \bar{\boldsymbol{r}}(W)\|^2. \tag{12}$$

The cross term vanishes because $\mathbb{E}_{\widetilde{P}}[\boldsymbol{r} - \bar{\boldsymbol{r}}(W) \mid W] = 0$. See equations 40 and 41.

Frozen-feature separation. In the scalar Gaussian witness of Appendix A, let c_1, c_2 be independent uniform signs, put $p = c_1 c_2$ and take $x_0 = \delta p + \xi$, where $\delta > 0$ and $\xi \sim \mathcal{N}(0, 1)$. The frozen representation exposes the interaction through $h_L = p$. With $\tau_\sigma = \sigma/[(1-\sigma)^2 + \sigma^2]$, the one-parameter refiner is $g_b = -b\tau_\sigma p$, whereas $\mathcal{A}$ contains raw-affine functions of (x_σ, c_1, c_2) with arbitrary σ-dependent coefficients.

Theorem 2 (Frozen-feature advantage)**.** *For an explicit $\Delta_{\text{aff}} > 0$ defined in Appendix A,*

$$\mathcal{R}_P(g_\delta) = \mathcal{R}_{\text{irr}}, \qquad \inf_{a\in\mathcal{A}} \mathcal{R}_P(a) = \mathcal{R}_{\text{irr}} + \Delta_{\text{aff}}.$$

If $\widehat{b}$ is ϵ_{opt}-suboptimal for $\mathcal{J}_\lambda(b) = \mathcal{R}_P(g_b) + \lambda\mathbb{E}_{\widetilde{P}} L_{\cos}(g_b, r)$, with $\lambda = 0.1$, then

$$2\lambda + \epsilon_{\text{opt}} < \Delta_{\text{aff}} \quad \Longrightarrow \quad \mathcal{R}_P(g_{\widehat{b}}) < \inf_{a\in\mathcal{A}} \mathcal{R}_P(a). \tag{13}$$

Matched-resource separation. Returning to the finite-dimensional space $\mathcal{X}$, we compare two runs from the same checkpoint $\boldsymbol{v}_0$: the refiner correction $F_{\text{r}}(\boldsymbol{\alpha}) = \boldsymbol{g}_{\boldsymbol{\phi}_0+\boldsymbol{\alpha}}(\mathcal{Z}(W))$ and continued base training $F_{\text{b}}(\boldsymbol{\alpha}) = \boldsymbol{v}_{\boldsymbol{\theta}_0+\boldsymbol{\alpha}}(W) - \boldsymbol{v}_0(W)$, with $F_j(0) = 0$. Under a common ledger $\mathcal{C}$, arm j takes $T_j = \lfloor C_{\text{train}}/c_j \rfloor$ gradient steps and returns $\boldsymbol{f}_{j,\eta}$. Write $\Delta_j = \sup_{\eta\in\mathcal{E}_j}[\mathcal{R}_P(0) - \mathcal{R}_P(\boldsymbol{f}_{j,\eta})]$, $\Phi_j = DF_j(0)$, $H_j = \Phi_j^*\Phi_j$ and $\boldsymbol{b}_j = \Phi_j^*\bar{\boldsymbol{r}}$.

Theorem 3 (Finite-ledger refiner separation)**.** *Under the local Jacobian, angular-gradient, radius and spectral-stability conditions of Appendix B.1, both armwise optima are attained and the actual reduction Δ_j differs from its tangent approximation Δ_j^{lin} by at most an explicit finite error e_j:*

$$|\Delta_j - \Delta_j^{\text{lin}}| \le e_j, \qquad \Delta_j^{\text{lin}} = \sup_{\eta\in\mathcal{E}_j} \boldsymbol{b}_j^* H_j^\dagger [I - (I - \eta H_j)^{2T_j}]\boldsymbol{b}_j.$$

Consequently,

$$\Delta_{\text{r}}^{\text{lin}} - e_{\text{r}} > \Delta_{\text{b}}^{\text{lin}} + e_{\text{b}} \Longrightarrow \min_{\eta\in\mathcal{E}_{\text{r}}} \mathcal{R}_P(\boldsymbol{f}_{\text{r},\eta}) < \min_{\eta\in\mathcal{E}_{\text{b}}} \mathcal{R}_P(\boldsymbol{f}_{\text{b},\eta}). \tag{14}$$

Own-law deployment. Write g_ϕ for the deployed correction $W \mapsto g_\phi(\mathcal{Z}(W))$. Let $\mathbf{X}^\kappa$ be the exact trajectory from the prescribed source under the field $v_0 + \kappa g_\phi$. For Σ drawn independently of the source and condition from a law q on $[0,1]$, set $Q_\kappa = \mathrm{Law}(\mathbf{X}^\kappa(\Sigma), \Sigma, \boldsymbol{C})$ and define $\mathcal{V}_Q(\kappa) = \mathbb{E}_Q\|\bar{\boldsymbol{r}} - \kappa g_\phi\|^2$. Here $\bar{\boldsymbol{r}}$ denotes the deployment version fixed in Appendix C.1.

Theorem 4 (Own-law velocity improvement)**.** *Under the deployment assumptions of Appendix C, let M be the first-order alignment margin after trajectory-shift cost and N the second-order correction and alignment-shift cost, as defined in Appendix C.3. Then, for every $\kappa \in [0, \bar{\kappa}]$,*

$$\mathcal{V}_{Q_\kappa}(\kappa) - \mathcal{V}_{Q_0}(0) \le -\kappa M + \kappa^2 N.$$

If $M > 0$ (hence $N > 0$), then

$$0 < \kappa \le \bar{\kappa}, \quad \kappa < \frac{M}{N} \quad \Longrightarrow \quad \mathcal{V}_{Q_\kappa}(\kappa) < \mathcal{V}_{Q_0}(0). \tag{15}$$

5 Experiments

We evaluate CorrFlow on a wide range of tasks, including class-conditional image generation, text-to-video generation and image-to-video generation. The image experiments test whether learned velocity corrections improve an established flow-matching model, while the video experiments examine visual quality, temporal consistency and physical plausibility. Red, orange and yellow highlight the best, second-best and third-best scores in each column, respectively.

5.1 Image Generation

Settings. We study class-conditional generation on ImageNet at 256×256 resolution with 1,000 classes. Our base model is built upon RAEv2 with a frozen DINOv3-L-based representation autoencoder and a pretrained diffusion transformer. Starting from its 80-epoch checkpoint, we jointly fine-tune the base transformer and train a randomly initialized refiner for another 80 epochs on the ImageNet training split, denoted by $80 + 80$ in Table 1. The refiner has 10 transformer blocks with width 768 and 12 attention heads, and reads hidden states from base transformer blocks 3, 11, 19 and 27, using one-based indexing. We retain the base model's training objective and train the refiner to predict the realized velocity residual with squared error and a cosine-direction term weighted by 0.1. Both networks use GMuon optimizer with a global batch size of 1,024. The evaluation settings are identical to those in Singh et al. (2026).

Table 1: Comparison on ImageNet-256 (Deng et al., 2009) generation. Representation Fréchet Distance (FD_r) is reported in three feature spaces.

Method	Epochs	Guidance	IS ↑	Representation Fréchet Distance (FD_r) ↓		
				ConvNeXt (Liu et al., 2022)	**DINOv2** (Oquab et al., 2023)	**CLIP** (Radford et al., 2021)
RAEv2 (Singh et al., 2026)	160	No	233.9	1.28	2.22	7.18
SelfFlow (Chefer et al., 2026)	160	No	146.4	4.81	7.08	12.13
Ours	80+80	No	234.0	1.24	2.21	7.14
SiT-XL/2 (Ma et al., 2024)	800	Yes	256.7	2.02	7.89	17.69
RAE-XL (Zheng et al., 2026a)	800	Yes	261.0	1.79	2.11	7.87
RAEv2 (Singh et al., 2026)	160	Yes	265.0	0.74	1.08	4.85
SelfFlow (Chefer et al., 2026)	160	Yes	146.8	4.72	6.98	12.16
Ours	80+80	Yes	**266.6**	**0.73**	**1.03**	**4.84**

Results. Table 1 compares CorrFlow with RAEv2 at a matched budget of 160 total epochs, because our $80 + 80$ schedule spends 80 epochs on the base checkpoint and 80 epochs on joint training. Under this budget, CorrFlow improves over RAEv2 on IS and on all three FD_r metrics, both with and without guidance, so the gain comes from the learned residual corrections rather than from a longer training schedule. With guidance, CorrFlow leads on every metric in the table, surpassing SiT-XL/2 and RAE-XL trained for 800 epochs. It also outperforms the evaluated SelfFlow variant in both guidance settings. Additional results are provided in Appendix E.1.

5.2 Text-to-Video Generation

Settings. We use Wan2.1-T2V-1.3B (Wan et al., 2025) as the base model and jointly adapt its transformer with the refiner on video–text pairs from WISA-80K (Wang et al., 2025) and Open-

Table 2: Physics-consistent text-to-video generation on VideoPhy2 (Bansal et al., 2026) and FinePhyEval (Pothiraj et al., 2026). Higher is better for all metrics.

Method	VideoPhy2						FinePhyEval			
	ALL			HARD			Object	Action	Physics	Overall
	PC	SA	Joint	PC	SA	Joint				
Wan 2.1-14B (Wan et al., 2025)	60.5	29.0	24.7	45.6	8.3	3.9	0.86	0.53	**0.46**	0.59
PhysRAG-5B (Cheng et al., 2026)	65.6	24.7	21.0	59.4	7.2	4.4	0.88	0.49	0.37	0.55
VideoREPA-5B (Zhang et al., 2025c)	**73.6**	25.8	23.6	**62.8**	6.7	5.0	0.82	0.37	0.30	0.47
Wan 2.1-1.3B (Wan et al., 2025)	57.8	30.0	24.8	36.7	**11.7**	5.6	0.84	0.49	0.36	0.54
DreamWorld-1.3B (Tan et al., 2026)	53.7	27.0	20.9	35.0	7.8	3.9	**0.90**	0.53	0.38	0.58
VPT-1.3B (Zheng et al., 2026b)	55.1	22.5	22.2	39.4	8.3	5.6	0.82	0.44	0.34	0.50
CORRFLOW (Ours)-1.3B	65.7	**31.3**	**27.0**	51.9	11.0	**6.9**	**0.90**	**0.55**	0.44	**0.60**

VidHD (Nan et al., 2025). The refiner has 10 transformer blocks with width 1,024 and 16 attention heads, and reads hidden states from base blocks 4, 12, 20 and 30, using one-based indexing. The final training stage uses 66,572 clips for five epochs with AdamW and a global batch size of 16, retaining the base flow-matching objective and a residual objective with cosine-direction weight 0.1. We generate 81-frame videos at 480×832 resolution with 50 Euler steps and evaluate visual quality and temporal consistency with VBench (Huang et al., 2024), and physical plausibility and prompt adherence with VideoPhy2 (Bansal et al., 2026) and FinePhyEval (Pothiraj et al., 2026). Appendix D.2 provides the training and evaluation details.

Table 3: VBench (Huang et al., 2024) comparison on selected metrics (0–100, higher is better).

Method	SC	TF	MS	AQ	IQ	OC	SR	TS	QS	Total
VideoREPA-5B (Zhang et al., 2025c)	95.47	98.23	98.54	57.19	59.80	67.56	51.85	22.71	79.96	76.67
PhysRAG-5B (Cheng et al., 2026)	96.31	98.71	96.93	56.80	65.56	80.50	77.33	23.18	83.07	80.08
Wan 2.1-1.3B (Wan et al., 2025)	91.83	99.17	96.51	57.24	59.14	76.09	62.37	22.23	79.81	76.93
VideoJAM-1.3B (Chefer et al., 2025)	91.51	99.13	96.05	58.41	62.68	79.22	66.17	24.03	81.18	78.76
DreamWorld-1.3B (Tan et al., 2026)	93.62	98.81	98.07	59.59	66.76	81.32	70.47	23.82	83.49	80.97
VPT-1.3B (Zheng et al., 2026b)	92.65	98.64	97.85	58.09	63.68	73.43	73.35	22.52	82.26	78.79
CORRFLOW (Ours)-1.3B	**96.68**	**99.81**	**98.85**	**61.65**	**68.66**	**82.81**	**77.88**	**24.05**	**84.69**	**81.86**

Results. Table 2 reports the VideoPhy2 and FinePhyEval results for physics-consistent text-to-video generation. CORRFLOW achieves the highest VideoPhy2 Joint scores on both ALL and HARD, increasing them from 24.8 to 27.0 and from 5.6 to 6.9, respectively, over Wan2.1-T2V-1.3B. It also achieves the highest FinePhyEval Overall score of 0.60, with improvements in object, action and physics scores over the same base model. On VBench (table 3), CORRFLOW leads the compared methods on Quality Score and Overall Score, reaching 84.69 and 81.86. Figure 3 shows the same advantage in the generated frames, where CORRFLOW carries each prompted interaction to its physical outcome, dropping the domino into the mug and landing the coin in the metal cup, while the baselines lose the object or drift away from the prompted action. These results show that the adapted model improves both general video quality and the joint satisfaction of semantic and physical requirements. Appendix E.2 analyzes the full metric breakdown.

5.3 IMAGE-TO-VIDEO GENERATION

Settings. We use Wan2.2-TI2V-5B (Wan et al., 2025) as the base model, with the same training data, objectives, optimization and five-epoch schedule as in Section 5.2. We enable first-frame image conditioning for 40% of training examples and use text-only conditioning for the remainder. The refiner has 13 transformer blocks with width 2,048 and 32 attention heads, reading hidden states from the same base blocks as in text-to-video generation. We evaluate on the 198 image–text pairs of Physics-IQ Verified (Rädsch et al., 2026; Motamed et al., 2026), generating 81-frame

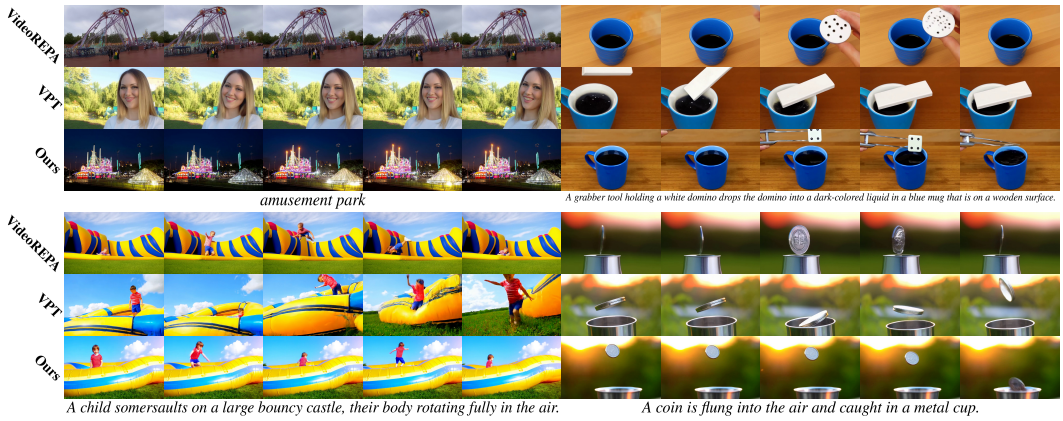

Figure 3: **Qualitative comparison on text-to-video generation.** Each scenario compares VideoREPA-5B, VPT-1.3B and CORRFLOW (Ours) under the same text prompt, with frames sampled at 0%, 25%, 50%, 75% and 100% of each generated video. The prompts are drawn from VBench (amusement park), FinePhyEval (a domino dropped into a mug) and VideoPhy2 (a somersault on a bouncy castle, a coin caught in a metal cup).

videos at 704×1248 resolution with the same sampling configuration as in text-to-video generation. Appendix D.3 details the image conditioning, training and evaluation.

Table 4: Physics-IQ Verified (Rädsch et al., 2026) results on the image-to-video track. All evaluations are without LLM usage. All scores are percentages, and higher is better for every metric.

Method	Spatial	Spatio-temporal	Weighted Spatial	MSE Score	Verified Score
P-Video-19B (Pruna AI, 2026)	38.60	16.40	22.90	23.30	25.30
Wan2.2-5B (Wan et al., 2025)	35.89	**26.44**	22.96	25.54	27.71
WMReward-5B (Yuan et al., 2026)	36.47	22.38	19.79	27.26	26.48
VLIPP-5B (Yang et al., 2025)	40.51	19.09	23.66	26.48	27.43
PhysRAG-5B (Cheng et al., 2026)	36.46	22.07	20.22	23.41	25.54
CORRFLOW (Ours)-5B	**43.43**	26.37	**24.74**	**29.24**	**30.95**

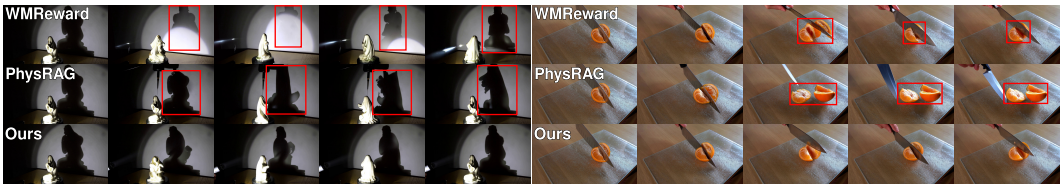

Figure 4: **Qualitative comparison on the Physics-IQ Verified image-to-video track.** Each scenario compares WMReward-5B, PhysRAG-5B and CORRFLOW (Ours), which share the Wan2.2-TI2V-5B base model and receive the same settings. Temporal order is from left to right. The left scenario shows a statuette's shadow under a moving light, and the right scenario shows a knife slicing a mandarin.

Results. Table 4 reports the Physics-IQ Verified results. CORRFLOW achieves the highest Verified Score of 30.95, and leads on Spatial, Weighted Spatial and MSE Score. These results indicate closer agreement with the ground-truth videos in motion coverage and frame appearance. Figure 4 illustrates shadow and object consistency, with the statuette's shadow tracking the moving light and the blade separating the mandarin into halves. Appendix E.3 provides the metric breakdown and qualitative analysis.

5.4 Ablation Study

We conduct a number of ablations to nanalyze the contribution of each component. All variants are built on WAN-2.1-T2V-1.3B (Wan et al., 2025) and follow the same training protocol. Specifically, each model is trained for five epochs on the same 1,000 samples from WISA-80K (Wang et al., 2025). We evaluate on a disjoint set of 50 WISA-80K samples selected to provide approximately uniform coverage over the physical categories. We report visual quality and consistency metrics from VBench, together with its aggregate Quality Score. Results are reported in Table 5.

Table 5: Component ablations (%, higher is better). Colors rank the four-decimal scores in Table 10, which also expands the metric names.

Method	**AQ**	**BC**	**IQ**	**OC**	**SC**	**QS**
w/o time-step	51.5	92.4	63.7	58.5	91.2	79.7
w/o self-attention	48.9	92.1	60.4	58.1	90.0	78.5
w/o cross-attention	48.5	91.4	60.7	58.2	90.2	78.5
w/o hidden-state MLP	49.3	92.4	63.6	58.2	90.9	79.3
w/o refiner	51.3	92.5	62.0	59.1	91.3	78.6
$\boldsymbol{v}$ input	50.4	92.4	63.7	60.2	91.7	79.8
w/o condition	51.5	92.7	61.2	58.1	91.0	79.8
Ours	**52.6**	**93.6**	**64.3**	**61.8**	**92.6**	**80.3**

The full model achieves the highest Quality Score and leads on all visual quality and consistency metrics. Removing cross-attention causes the largest score drop, closely followed by self-attention, highlighting the value of context and spatiotemporal interactions in the base model's prediction. Removing either the refiner or hidden-state MLP reduces Quality Score and Overall Consistency, supporting refinement and fusion of hidden states across depths. The remaining ablations produce smaller but consistent score decreases. Removing noise-level conditioning or replacing the clean prediction $\hat{\mathbf{x}}_0$ with the velocity prediction $\boldsymbol{v}_{\boldsymbol{\theta}}$ lowers aggregate performance, supporting the value of noise-level information and explicit clean-sample estimates. Removing text conditioning lowers Quality Score and notably weakens Overall Consistency, suggesting its role in semantic alignment.

6 Conclusion

We introduced CorrFlow, a self-correcting flow-matching framework that learns velocity residuals from the base model's clean prediction, conditioning and hidden states. A lightweight refiner corrects the velocity within the sampling loop. Our theory identifies conditions under which feature reuse improves residual prediction, refiner training outperforms continued base-model training at matched compute, and the deployed correction lowers own-law velocity error. We demonstrate that CorrFlow achieves state-of-the-art performance across multiple benchmarks including image generation, text-to-video generation, and image-to-video generation. We hope CorrFlow will be a useful tool for the community to advance research in flow-matching-based generative modeling.

AI USE STATEMENT

We discussed ideas with large language models while developing the theoretical derivations. The derivations themselves were completed and verified by the authors. Beyond this, we did not use generative AI tools for any other tasks for which the ICLR policy requires disclosure, including data generation, hypothesis formulation, experimental design, method implementation, translation, data processing, and result interpretation. We used large language models mainly for tasks with recommended disclosure: editing the manuscript for readability, searching for relevant prior work, and drafting parts of the paper. The authors revised every AI-assisted passage against our own method, code, and results, and verified every AI-suggested reference against the original publication. We take full responsibility for the final content of this work, including all AI-assisted text.

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A FROZEN-FEATURE ANALYSIS

This appendix establishes an exact population-risk separation for a linear readout of a frozen nonlinear feature relative to predictors that are affine in the raw inputs.

A.1 SETTING AND ASSUMPTIONS

We work in one active coordinate of the sample space. Samples, velocities, residuals and hidden readouts in this construction are scalar and use ordinary symbols. The conditioning vector is bold. Let $\boldsymbol{c} = (c_1, c_2)$, write $p := \psi(\boldsymbol{c}) := c_1 c_2$ and fix $\delta > 0$. For $\sigma \in [0, 1]$, define

$$s_\sigma^2 := (1-\sigma)^2 + \sigma^2, \qquad \ell_\sigma := \frac{2\sigma - 1}{s_\sigma^2}, \qquad \tau_\sigma := \frac{\sigma}{s_\sigma^2}.$$

Assumption 1 (Conditional-Gaussian XOR witness)**.** *The variables c_1, c_2 are independent and uniform on $\{-1, 1\}$. The variables $\xi, \varepsilon \sim \mathcal{N}(0, 1)$ are independent of each other and of $(\boldsymbol{c}, \sigma)$, and*

$$x_0 := \delta p + \xi, \qquad x_\sigma := (1-\sigma)x_0 + \sigma\varepsilon, \qquad u := \varepsilon - x_0.$$

Assumption 2 (Independent Noise Level)**.** *The variable $\sigma \sim q$ is independent of $(\boldsymbol{c}, \xi, \varepsilon)$, where q is a Borel probability measure with topological support exactly $[0, 1]$.*

Let $(\Omega, \mathscr{F}, \mathbb{P})$ be the underlying probability space carrying these variables. Write $W := (x_\sigma, \sigma, \boldsymbol{c})$ and let $P_W := \mathbb{P} \circ W^{-1}$ be its law. The frozen base velocity, its one-step clean prediction from Equation 5 and the realized residual label are

$$v_{\boldsymbol{\theta}}(W) := \ell_\sigma x_\sigma, \qquad \widehat{x}_0(W) := x_\sigma - \sigma v_{\boldsymbol{\theta}}(W), \qquad r := u - v_{\boldsymbol{\theta}}(W).$$

Assumption 3 (Frozen base and product feature)**.** *The base is frozen with velocity $v_{\boldsymbol{\theta}}(W) = \ell_\sigma x_\sigma$. Its captured representation is the deterministic map $\Phi(W) := (\widehat{x}_0(W), h_L(W), \sigma, \boldsymbol{c})$ with hidden coordinate $h_L(W) = c_1 c_2$. The base velocity readout omits this coordinate, while the scalar AdaLN-modulated class $\mathcal{G}$ may read it.*

Define

$$u^\star(W) := \mathbb{E}[u \mid W], \qquad \bar{r}(W) := \mathbb{E}[r \mid W] = u^\star(W) - v_{\boldsymbol{\theta}}(W).$$

All expectations below are with respect to $\mathbb{P}$. For every square-integrable W-measurable predictor f,

$$\mathcal{R}(f) := \mathbb{E}[(f(W) - r)^2], \qquad \mathcal{R}_{\mathrm{irr}} := \mathbb{E}[(r - \bar{r}(W))^2].$$

In the notation of Section 4, $P = P_W$ and $\widetilde{P} = \mathbb{P} \circ (W, r)^{-1}$. In this scalar specialization, the frozen checkpoint $\boldsymbol{v}_0$ corresponds to $v_{\boldsymbol{\theta}}$ and $\mathcal{R}_P = \mathcal{R}$.
The one-parameter hidden-feature class, viewed as induced predictors on W, is

$$\mathcal{G} := \{g_b : b \in \mathbb{R}\}, \qquad g_b(W) := -b\tau_\sigma h_L(W).$$

Each g_b uses only $\Phi(W)$. As an oracle enlargement of the intended raw linear readouts, let

$$\mathcal{A} := \left\{ a_0(\sigma) + a_x(\sigma)x_\sigma + a_1(\sigma)c_1 + a_2(\sigma)c_2 : \begin{array}{l} a_0, a_x, a_1, a_2 \text{ are Borel measurable,} \\ \text{and the induced predictor lies in } L^2(P_W) \end{array} \right\}.$$

The intended one-scalar raw-input comparator families at the same trainable-readout budget are subclasses of $\mathcal{A}$. For nested-class comparisons we use the convention $\inf \varnothing = +\infty$.

For scalars a, b, define the zero-stabilized cosine loss by

$$\cos_0(a, b) := \begin{cases} ab/(|a||b|), & ab \neq 0, \\ 0, & ab = 0, \end{cases} \qquad L_{\cos}(a, b) := 1 - \cos_0(a, b).$$

For $\lambda = 0.1$, set

$$\mathcal{J}_\lambda(b) := \mathbb{E}\left[(g_b(W) - r)^2 + \lambda L_{\cos}(g_b(W), r)\right].$$

Assumption 4 (Cosine-regularized population fit)**.** *For a stated tolerance $\epsilon_{\mathrm{opt}} \geq 0$, the trained coefficient $\widehat{b} \in \mathbb{R}$ satisfies*

$$\mathcal{J}_\lambda(\widehat{b}) \leq \inf_{b \in \mathbb{R}} \mathcal{J}_\lambda(b) + \epsilon_{\mathrm{opt}}.$$

Define the structural constants

$$S := \mathbb{E}_{\sigma \sim q}[\tau_\sigma^2], \qquad \Delta_{\mathrm{aff}} := \delta^2 \mathbb{E}_{\sigma \sim q}\left[\tau_\sigma^2 \frac{s_\sigma^2}{s_\sigma^2 + (1-\sigma)^2\delta^2}\right].$$

A.2 MAIN RESULT

Theorem 5 (Detailed frozen-feature advantage)**.** *Under Assumptions 1– 4, the following statements hold.*

1. *For every $\sigma < 1$,*

$$x_\sigma = \frac{s_\sigma^2}{1-\sigma}\widehat{x}_0,$$

whereas $\widehat{x}_0 = 0$ at $\sigma = 1$. On the full interval,

$$\widehat{x}_0 - x_0 = \sigma r.$$

Thus $r = (\widehat{x}_0 - x_0)/\sigma$ for $\sigma > 0$, with the direct continuous endpoint values $r = \varepsilon$ at $\sigma = 0$ and $r = -x_0$ at $\sigma = 1$. Moreover,

$$u^\star(W) = \ell_\sigma x_\sigma - \tau_\sigma \delta p, \qquad \bar{r}(W) = -\tau_\sigma \delta p.$$

2. *The irreducible risk and structural constants satisfy*

$$\mathcal{R}_{\mathrm{irr}} = \mathbb{E}_{\sigma\sim q}[s_\sigma^{-2}] \in [1,2], \qquad 0 < S < \infty, \qquad 0 < \Delta_{\mathrm{aff}} < \infty.$$

For every $b \in \mathbb{R}$,

$$\mathcal{R}(g_b) = \mathcal{R}_{\mathrm{irr}} + S(b-\delta)^2, \qquad \inf_{a\in\mathcal{A}} \mathcal{R}(a) = \mathcal{R}_{\mathrm{irr}} + \Delta_{\mathrm{aff}}.$$

In particular, $g_\delta = \bar{r}$ and $\mathcal{R}(g_\delta) = \mathcal{R}_{\mathrm{irr}}$.

3. *The fitted coefficient satisfies*

$$S(\widehat{b} - \delta)^2 \le 2\lambda + \epsilon_{\mathrm{opt}}.$$

If the static scalar condition

$$2\lambda + \epsilon_{\mathrm{opt}} < \Delta_{\mathrm{aff}}$$

holds, then

$$\mathcal{R}(g_{\widehat{b}}) < \inf_{a\in\mathcal{A}} \mathcal{R}(a).$$

For every intended equal-budget raw comparator class $\mathcal{B} \subseteq \mathcal{A}$, the same condition also gives

$$\mathcal{R}(g_{\widehat{b}}) < \inf_{f\in\mathcal{B}} \mathcal{R}(f).$$

4. *For every finite $d \ge 1$, the embedding $\iota_d(z) = (z, 0, \ldots, 0)$ preserves the feature-risk curve, the raw-affine infimum, the cosine-regularized objective and the strict trained-risk margin.*

These identities give the hidden-feature refiner's population $L^2(\mathbb{P})$ risk and its margin relative to the specified raw-affine classes. The parameters δ, q, λ and ϵ_{opt} determine the feature curvature S, the raw-affine gap Δ_{aff} and the additive excess-risk bound.

The product feature represents the conditional Bayes residual exactly. The noise law q weights the risk across noise levels, with full support ensuring positive feature curvature S and raw-affine gap Δ_{aff}. The cosine weight λ and optimization tolerance ϵ_{opt} bound the fitted predictor's excess risk. The strict scalar condition places this bound below the raw-affine gap, certifying the trained refiner's advantage.

Theorem 5 is the detailed version of the main-text Theorem 2.

A.3 PROOF

Proof. We prove the scalar conclusions in four steps. The finite-dimensional preservation statement is verified in the embedding paragraph below.

Step 1: Residual interface and conditional Bayes residual. Since

$$s_\sigma^2 = 2\left(\sigma - \frac{1}{2}\right)^2 + \frac{1}{2} \geq \frac{1}{2},$$

all denominators below are nonzero. Moreover,

$$1 - \sigma\ell_\sigma = \frac{s_\sigma^2 - \sigma(2\sigma - 1)}{s_\sigma^2} = \frac{1-\sigma}{s_\sigma^2},$$

which proves the displayed clean-prediction formula and its inverse for $\sigma < 1$:

$$\widehat{x}_0 = \frac{1-\sigma}{s_\sigma^2} x_\sigma, \qquad x_\sigma = \frac{s_\sigma^2}{1-\sigma}\widehat{x}_0.$$

Direct substitution gives $\widehat{x}_0 = x_0$ at $\sigma = 0$ and $\widehat{x}_0 = 0$ at $\sigma = 1$. The interpolation identity also gives, without division,

$$\widehat{x}_0 - x_0 = x_\sigma - \sigma v_{\boldsymbol{\theta}} - x_0 = \sigma(\varepsilon - x_0 - v_{\boldsymbol{\theta}}) = \sigma r.$$

Consequently, $r = (\widehat{x}_0 - x_0)/\sigma$ for every $\sigma > 0$. The direct residual definition gives $r = \varepsilon$ at zero and $r = -x_0$ at one, which agree with the continuous endpoint extension.

To compute the conditional mean, center the observation as

$$Y_\sigma := x_\sigma - (1-\sigma)\delta p = (1-\sigma)\xi + \sigma\varepsilon$$

and introduce its orthogonal complement

$$N_\sigma := (1-\sigma)\varepsilon - \sigma\xi.$$

Since p is measurable with respect to $\boldsymbol{c}$ and

$$Y_\sigma = x_\sigma - (1-\sigma)\delta p, \qquad x_\sigma = Y_\sigma + (1-\sigma)\delta p,$$

the tuples $W = (x_\sigma, \sigma, \boldsymbol{c})$ and $(Y_\sigma, \sigma, \boldsymbol{c})$ generate the same sigma-field. Conditional on $(\sigma, \boldsymbol{c})$, the vector $(Y_\sigma/s_\sigma, N_\sigma/s_\sigma)$ is standard Gaussian with independent coordinates, because it is obtained from (ξ, ε) by the orthogonal transformation

$$\begin{pmatrix} Y_\sigma/s_\sigma \\ N_\sigma/s_\sigma \end{pmatrix} = \frac{1}{s_\sigma}\begin{pmatrix} 1-\sigma & \sigma \\ -\sigma & 1-\sigma \end{pmatrix}\begin{pmatrix} \xi \\ \varepsilon \end{pmatrix}.$$

Because (ξ, ε) is independent of $(\sigma, \boldsymbol{c})$, the fixed-t Gaussian regression above defines a conditional kernel for every value of $(\sigma, \boldsymbol{c})$. The displayed coefficient functions are Borel and integrable because $s_\sigma^2 \geq 1/2$. Since W and $(Y_\sigma, \sigma, \boldsymbol{c})$ generate the same sigma-field, these kernels give the Borel conditional versions

$$\mathbb{E}[\xi \mid W] = \frac{1-\sigma}{s_\sigma^2} Y_\sigma, \qquad \mathbb{E}[\varepsilon \mid W] = \frac{\sigma}{s_\sigma^2} Y_\sigma.$$

Using $u = \varepsilon - \xi - \delta p$ and $1 + (1-\sigma)\ell_\sigma = \tau_\sigma$ gives

$$u^\star(W) = \ell_\sigma x_\sigma - \tau_\sigma \delta p, \qquad \bar{r}(W) = u^\star(W) - v_{\boldsymbol{\theta}}(W) = -\tau_\sigma \delta p.$$

Moreover, conditional on $(\sigma, \boldsymbol{c})$, the normalized coordinate N_σ/s_σ is independent of Y_σ/s_σ. Conditioning additionally on W, equivalently on $(Y_\sigma, \sigma, \boldsymbol{c})$, therefore leaves its Gaussian law unchanged. The same calculation yields

$$r - \bar{r}(W) = \frac{N_\sigma}{s_\sigma^2}, \qquad (r - \bar{r}(W)) \mid W \sim \mathcal{N}\big(0, s_\sigma^{-2}\big).$$

Therefore

$$\mathcal{R}_{\mathrm{irr}} = \mathbb{E}_{\sigma\sim q}[s_\sigma^{-2}]. \tag{16}$$

Since $s_\sigma^2 \in [1/2, 1]$, this quantity belongs to $[1, 2]$.

Step 2: Same-target decomposition and feature oracle. The following decomposition holds on any probability space with $r \in L^2(\mathbb{P})$ and $f(W) \in L^2(\mathbb{P})$. Set $\bar{r}(W) := \mathbb{E}[r \mid W]$ and $\eta := r - \bar{r}(W)$. Conditional Jensen's inequality gives

$$\mathbb{E}[\bar{r}(W)^2] \le \mathbb{E}[r^2].$$

Thus η and $f(W) - \bar{r}(W)$ belong to L^2, and their product is integrable by Cauchy–Schwarz. The defining orthogonality property of conditional expectation gives

$$\mathbb{E}[(f(W) - \bar{r}(W))\eta] = \mathbb{E}[(f(W) - \bar{r}(W))\mathbb{E}[\eta \mid W]] = 0.$$

Expanding the square therefore yields

$$\mathcal{R}(f) = \mathcal{R}_{\mathrm{irr}} + \mathbb{E}[(f(W) - \bar{r}(W))^2].$$

This also proves the scalar case of equation 12. For a finite-dimensional $\mathcal{X}$, choose an orthonormal basis and apply the same identity to each coordinate of $\boldsymbol{r}$ and $\boldsymbol{f}(W)$. Conditional expectation acts coordinatewise, and summing the coordinate identities gives the squared-norm identity in equation 12. Returning to the present Gaussian witness, since $p^2 = 1$ and $g_b - \bar{r} = (\delta - b)\tau_\sigma p$, we obtain

$$\mathcal{R}(g_b) = \mathcal{R}_{\mathrm{irr}} + S(b - \delta)^2.$$

In particular, $g_\delta = \bar{r}$ pointwise for the chosen Borel version. The bound $0 \le \tau_\sigma \le 2$ gives $S \le 4$. Since τ_σ^2 is continuous and strictly positive on $(0, 1]$, the full support of q gives $S > 0$. More explicitly, for $K = [1/2, 3/4]$, one has $q(K) > 0$, $\tau_\sigma^2 \ge 1/4$ on K and hence

$$S \ge \frac{1}{4} q(K) > 0.$$

Step 3: Exact raw-affine projection. For each $t \in [0, 1]$, let $\mathbb{E}_t$ denote expectation over $(\boldsymbol{c}, \xi, \varepsilon)$ with the noise level fixed at $\sigma = t$. By independence of σ, every nonnegative or integrable Borel function H satisfies

$$\mathbb{E}[H] = \int_{[0,1]} \mathbb{E}_t[H]\, q(dt).$$

Fix $t \in [0, 1]$, put $m_t := 1 - t$ and define

$$V_t := s_t^2 + m_t^2 \delta^2, \qquad d_t := \delta^2 \tau_t^2 \frac{s_t^2}{V_t}.$$

Under the fixed-noise law, the Gram matrix of $(1, c_1, c_2, x_t)$ is diagonal with diagonal $(1, 1, 1, V_t)$, because

$$\mathbb{E}_t[x_t] = 0, \quad \mathbb{E}_t[x_t^2] = V_t, \quad \mathbb{E}_t[x_t c_1] = \mathbb{E}_t[x_t c_2] = \mathbb{E}_t[c_1 c_2] = 0.$$

The target-design moments are

$$\mathbb{E}_t[\bar{r}] = \mathbb{E}_t[\bar{r} c_1] = \mathbb{E}_t[\bar{r} c_2] = 0, \qquad \mathbb{E}_t[\bar{r} x_t] = -m_t \tau_t \delta^2, \qquad \mathbb{E}_t[\bar{r}^2] = \tau_t^2 \delta^2.$$

Consequently, for $A_\alpha := \alpha_0 + \alpha_x x_t + \alpha_1 c_1 + \alpha_2 c_2$, completion of the square gives

$$\mathbb{E}_t[(A_\alpha - \bar{r})^2] = \alpha_0^2 + \alpha_1^2 + \alpha_2^2 + V_t \left(\alpha_x + \frac{m_t \tau_t \delta^2}{V_t} \right)^2 + d_t.$$

This identity also holds at $t = 0, 1$, where $d_0 = 0$ and $d_1 = \delta^2$. The pointwise minimizer is therefore

$$a_0^\star = a_1^\star = a_2^\star = 0, \qquad a_x^\star(t) = -\frac{m_t \tau_t \delta^2}{V_t}.$$

Equivalently, let

$$a_t^\star := -\frac{m_t \tau_t \delta^2}{V_t} x_t, \qquad e_t^\star := \bar{r} - a_t^\star.$$

The displayed moment identities give

$$\mathbb{E}_t[e_t^\star] = \mathbb{E}_t[e_t^\star c_1] = \mathbb{E}_t[e_t^\star c_2] = \mathbb{E}_t[e_t^\star x_t] = 0.$$

Thus $a_t^\star$ is the conditional L^2 projection of $\bar{r}$ onto the raw-affine design, and

$$\mathbb{E}_t[(e_t^\star)^2] = d_t.$$

Because $V_t \geq s_t^2 \geq 1/2$, these coefficients are continuous and bounded on $[0,1]$. Moreover,

$$\sup_{t\in[0,1]} \mathbb{E}_t[x_t^2] = \sup_{t\in[0,1]} V_t \leq 1 + \delta^2.$$

Thus the induced predictor $a^\star(W) = a_x^\star(\sigma)x_\sigma$ is square-integrable and belongs to $\mathcal{A}$. For every $a \in \mathcal{A}$ with coefficient functions a_0, a_x, a_1, a_2, the same diagonal Gram calculation applied to $a(W)^2$ gives

$$\int_{[0,1]} \left(a_0(t)^2 + a_1(t)^2 + a_2(t)^2 + V_t a_x(t)^2\right) q(dt) = \mathbb{E}[a(W)^2] < \infty.$$

Hence the conditional square-completion identity may be integrated with respect to $q(dt)$, giving

$$\mathbb{E}[(a(W) - \bar{r}(W))^2] = \int_{[0,1]} \Bigg[a_0(t)^2 + a_1(t)^2 + a_2(t)^2 + V_t \left(a_x(t) + \frac{(1-t)\tau_t\delta^2}{V_t} \right)^2 + d_t \Bigg] q(dt).$$

Consequently,

$$\mathbb{E}[(a(W) - \bar{r}(W))^2] \geq \int_{[0,1]} d_t \, q(dt).$$

The coefficient choice $a^\star$ makes every squared term vanish for all $t \in [0,1]$, including both endpoints, so equality is attained. Therefore,

$$\inf_{a\in\mathcal{A}} \mathbb{E}[(a(W) - \bar{r}(W))^2] = \mathbb{E}_{t\sim q}[d_t] = \Delta_{\text{aff}}.$$

Applying the same-target decomposition gives

$$\inf_{a\in\mathcal{A}} \mathcal{R}(a) = \mathcal{R}_{\text{irr}} + \Delta_{\text{aff}}.$$

The bounds $0 \leq d_t \leq \delta^2\tau_t^2 \leq 4\delta^2$ prove finiteness. Moreover, d_t is continuous and strictly positive for every $t > 0$. Full support of q therefore gives $\Delta_{\text{aff}} > 0$. Explicitly, on $K = [1/2, 3/4]$,

$$d_t \geq \frac{\delta^2}{2(4+\delta^2)}, \qquad \Delta_{\text{aff}} \geq \frac{\delta^2}{2(4+\delta^2)} q(K) > 0.$$

Equivalently, define the squared conditional correlation

$$\rho_t^2 := \text{Corr}_t(p, x_t)^2 = \frac{m_t^2\delta^2}{V_t}.$$

The conditional gap satisfies

$$d_t = \text{Var}_t(\bar{r})(1 - \rho_t^2),$$

so it is exactly the product-feature energy unexplained by the raw-affine projection. The raw-affine projection explains the fraction ρ_t^2 of the conditional Bayes-residual energy, while the hidden product feature represents the target exactly.

Step 4: Transfer the fitted objective to strict risk separation. The zero-stabilized cosine loss is measurable and lies in $[0,2]$. Therefore, for every b,

$$\mathcal{R}(g_b) \leq \mathcal{J}_\lambda(b) \leq \mathcal{R}(g_b) + 2\lambda.$$

Using Assumption 4 and comparing with the oracle coefficient $b = \delta$ gives

$$\begin{aligned} \mathcal{R}(g_{\widehat{b}}) &\leq \mathcal{J}_\lambda(\widehat{b}) \\ &\leq \inf_b \mathcal{J}_\lambda(b) + \epsilon_{\text{opt}} \\ &\leq \mathcal{J}_\lambda(\delta) + \epsilon_{\text{opt}} \\ &\leq \mathcal{R}_{\text{irr}} + 2\lambda + \epsilon_{\text{opt}}. \end{aligned}$$

Combining this bound with the exact feature-risk curve yields

$$S(\widehat{b}-\delta)^2 \le 2\lambda+\epsilon_{\mathrm{opt}}, \qquad |\widehat{b}-\delta| \le \sqrt{\frac{2\lambda+\epsilon_{\mathrm{opt}}}{S}}. \tag{17}$$

For every b the exact class margin is

$$\inf_{a\in\mathcal{A}} \mathcal{R}(a) - \mathcal{R}(g_b) = \Delta_{\mathrm{aff}} - S(b-\delta)^2. \tag{18}$$

Since $S>0$ and $\Delta_{\mathrm{aff}}>0$, this identity gives the exact coefficient interval for strict advantage:

$$\mathcal{R}(g_b) < \inf_{a\in\mathcal{A}} \mathcal{R}(a) \quad\Longleftrightarrow\quad |b-\delta| < \sqrt{\frac{\Delta_{\mathrm{aff}}}{S}}. \tag{19}$$

The condition $2\lambda+\epsilon_{\mathrm{opt}} < \Delta_{\mathrm{aff}}$ places the fitted coefficient in this interval by equation 17. Evaluating at $\widehat{b}$ proves

$$\inf_{a\in\mathcal{A}} \mathcal{R}(a) - \mathcal{R}(g_{\widehat{b}}) \ge \Delta_{\mathrm{aff}} - (2\lambda+\epsilon_{\mathrm{opt}}). \tag{20}$$

Under the stated scalar condition this margin is strictly positive. If $\mathcal{B}\subseteq\mathcal{A}$, then $\inf_{a\in\mathcal{A}}\mathcal{R}(a) \le \inf_{f\in\mathcal{B}}\mathcal{R}(f)$, which proves the final claim. The fourth conclusion follows from the finite-dimensional embedding argument below. □

A.4 Useful specializations

Uniform noise law. If q is uniform on $[0,1]$, use

$$z := 2\sigma-1, \qquad s_\sigma^2 = \frac{1+z^2}{2}, \qquad d\sigma = \frac{1}{2}dz.$$

Then

$$\mathcal{R}_{\mathrm{irr}} = \int_{-1}^{1} \frac{1}{1+z^2}\,dz = \frac{\pi}{2},$$

while the odd part of $(z+1)^2$ integrates to zero and

$$S = \int_{-1}^{1} \frac{(z+1)^2}{2(1+z^2)^2}\,dz = \frac{1}{2}\int_{-1}^{1}\frac{1}{1+z^2}\,dz = \frac{\pi}{4}.$$

Thus

$$\mathcal{R}_{\mathrm{irr}} = \frac{\pi}{2}, \qquad S=\frac{\pi}{4}, \qquad \mathcal{R}(g_b) = \frac{\pi}{2} + \frac{\pi}{4}(b-\delta)^2. \tag{21}$$

Consequently,

$$|\widehat{b}-\delta| \le \frac{2}{\sqrt{\pi}}\sqrt{2\lambda+\epsilon_{\mathrm{opt}}}.$$

Finite-dimensional embedding. The witness uses one active sample coordinate so that every source of the risk gap is visible. Fix a finite $d\ge 1$ and define

$$\iota_d : \mathbb{R}\to\mathbb{R}^d, \qquad \iota_d(z) := (z,0,\dots,0),$$

$$\boldsymbol{r}^{(d)} := \iota_d(r), \qquad \bar{\boldsymbol{r}}^{(d)} := \iota_d(\bar{r}).$$

For a scalar predictor f, write $\boldsymbol{f}^{(d)} := \iota_d \circ f$ and set

$$\mathcal{R}_d(\boldsymbol{f}^{(d)}) := \mathbb{E}\Big[\|\boldsymbol{f}^{(d)}(W) - \boldsymbol{r}^{(d)}\|_2^2\Big].$$

The embedded classes are

$$\mathcal{G}_d := \{\boldsymbol{g}_b^{(d)} : b\in\mathbb{R}\}, \qquad \mathcal{A}_d := \{\boldsymbol{a}^{(d)} : a\in\mathcal{A}\}.$$

Since ι_d preserves inner products, norms and pointwise distances,

$$\mathbb{E}[\boldsymbol{r}^{(d)} \mid W] = \bar{\boldsymbol{r}}^{(d)}, \qquad \mathcal{R}_d(\boldsymbol{g}_b^{(d)}) = \mathcal{R}(g_b), \qquad \mathcal{R}_d(\boldsymbol{a}^{(d)}) = \mathcal{R}(a).$$

The map $a \mapsto \boldsymbol{a}^{(d)}$ is a bijection from $\mathcal{A}$ to $\mathcal{A}_d$. Therefore

$$\inf_{\boldsymbol{a}^{(d)} \in \mathcal{A}_d} \mathcal{R}_d(\boldsymbol{a}^{(d)}) = \mathcal{R}_{\mathrm{irr}} + \Delta_{\mathrm{aff}}, \qquad \boldsymbol{g}_\delta^{(d)} = \bar{\boldsymbol{r}}^{(d)}.$$

For $\boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^d$, define

$$\cos_{0,d}(\boldsymbol{y}, \boldsymbol{z}) := \begin{cases} \langle \boldsymbol{y}, \boldsymbol{z} \rangle / (\|\boldsymbol{y}\|_2 \|\boldsymbol{z}\|_2), & \|\boldsymbol{y}\|_2 \|\boldsymbol{z}\|_2 > 0, \\ 0, & \|\boldsymbol{y}\|_2 \|\boldsymbol{z}\|_2 = 0, \end{cases}$$

and $L_{\cos,d} := 1 - \cos_{0,d}$. Define

$$\mathcal{J}_{\lambda,d}(b) := \mathbb{E}\Big[\|\boldsymbol{g}_b^{(d)}(W) - \boldsymbol{r}^{(d)}\|_2^2 + \lambda L_{\cos,d}(\boldsymbol{g}_b^{(d)}(W), \boldsymbol{r}^{(d)}) \Big].$$

The same isometry gives $\mathcal{J}_{\lambda,d}(b) = \mathcal{J}_\lambda(b)$ for every b. Consequently, if $2\lambda + \epsilon_{\mathrm{opt}} < \Delta_{\mathrm{aff}}$, then

$$\mathcal{R}_d(\boldsymbol{g}_{\widehat{b}}^{(d)}) < \inf_{\boldsymbol{a}^{(d)} \in \mathcal{A}_d} \mathcal{R}_d(\boldsymbol{a}^{(d)}),$$

with margin at least $\Delta_{\mathrm{aff}} - (2\lambda + \epsilon_{\mathrm{opt}})$, independently of d. This is an isometric embedding of the scalar witness, not a claim about a general multidimensional data model.

B Matched-Resource Comparison: Proof

This section proves that a refiner's tangent risk-reduction advantage transfers to its trained output when the advantage exceeds the two perturbation bounds, as stated in Theorem 3. Unsubscripted norms of random correction maps are those of $L^2(P; \mathcal{X})$. Norms inside an expectation are sample-space norms, while parameter and operator norms have their usual finite-dimensional meanings. Linear operators retain ordinary uppercase symbols, while parameter vectors and vector-valued fields are bold.

B.1 Full setting and local certificate

Let $\mathcal{X}$ be a finite-dimensional real Hilbert space. Write $W = (\mathbf{X}_\Sigma, \Sigma, \boldsymbol{C})$ for the current noisy sample, noise level and conditioning vector, and let P be its law. Let $\boldsymbol{U}$ denote the realized interpolation velocity. At a fixed base checkpoint $\boldsymbol{v}_0$, define

$$\boldsymbol{r} = \boldsymbol{U} - \boldsymbol{v}_0(W), \qquad \widetilde{P} = \mathrm{Law}(W, \boldsymbol{r}), \qquad \bar{\boldsymbol{r}}(W) = \mathbb{E}_{\widetilde{P}}[\boldsymbol{r} \mid W], \qquad s = \|\boldsymbol{r}\|_{L^2(\widetilde{P}; \mathcal{X})} < \infty. \tag{22}$$

For $\boldsymbol{f} \in L^2(P; \mathcal{X})$, set

$$\mathcal{R}_P(\boldsymbol{f}) = \mathbb{E}_{\widetilde{P}} \|\boldsymbol{r} - \boldsymbol{f}(W)\|^2. \tag{23}$$

The refiner and continued-base correction families are

$$F_{\mathrm{r}}(\boldsymbol{\alpha})(W) = \boldsymbol{g}_{\boldsymbol{\phi}_0 + \boldsymbol{\alpha}}(\mathcal{Z}(W)), \qquad F_{\mathrm{b}}(\boldsymbol{\alpha})(W) = \boldsymbol{v}_{\boldsymbol{\theta}_0 + \boldsymbol{\alpha}}(W) - \boldsymbol{v}_0(W), \tag{24}$$

where $F_j(0) = 0$ and $\mathcal{Z}(W)$ is computed only from the current state and frozen base computation, as in Section 3.2.

A componentwise ledger $\mathcal{C} = (C_{\mathrm{train}}, C_{\mathrm{state}}, C_{\mathrm{deploy}})$ is fixed before either run. If one update of arm $j \in \{\mathrm{r}, \mathrm{b}\}$ costs $c_j > 0$, its charged horizon is

$$T_j = \left\lfloor \frac{C_{\mathrm{train}}}{c_j} \right\rfloor. \tag{25}$$

Both runs must fit the declared optimizer-state and inference budgets. Their population objectives are

$$L_{\mathrm{b}}(\boldsymbol{\alpha}) = \tfrac{1}{2} \mathcal{R}_P(F_{\mathrm{b}}(\boldsymbol{\alpha})), \qquad L_{\mathrm{r}}(\boldsymbol{\alpha}) = \tfrac{1}{2} \mathcal{R}_P(F_{\mathrm{r}}(\boldsymbol{\alpha})) + \lambda_\angle \mathcal{A}(\boldsymbol{\alpha}), \quad \lambda_\angle = 0.1, \tag{26}$$

where $\mathcal{A}$ is the population angular penalty induced by the implemented thresholded cosine loss. For a candidate step $\eta > 0$, define

$$\boldsymbol{\alpha}_{j,0}^\eta = 0, \qquad \boldsymbol{\alpha}_{j,t+1}^\eta = \boldsymbol{\alpha}_{j,t}^\eta - \eta \nabla L_j(\boldsymbol{\alpha}_{j,t}^\eta), \qquad \boldsymbol{f}_{j,\eta} = F_j(\boldsymbol{\alpha}_{j,T_j}^\eta). \tag{27}$$

At the checkpoint, let

$$\Phi_j = J_j(0), \qquad H_j = \Phi_j^* \Phi_j, \qquad \boldsymbol{b}_j = \Phi_j^* \bar{\boldsymbol{r}}, \tag{28}$$

where $J_j(\boldsymbol{\alpha}) = DF_j(\boldsymbol{\alpha})$. The zero-initialized tangent recursion is

$$\boldsymbol{\beta}_{j,t+1}^\eta = (I - \eta H_j) \boldsymbol{\beta}_{j,t}^\eta + \eta \boldsymbol{b}_j, \qquad \boldsymbol{\beta}_{j,0}^\eta = 0. \tag{29}$$

Assumption 5 (Finite-budget local certificate)**.** *For each arm there are finite $R_j, K_j, \Lambda_j \geq 0$ such that F_j is continuously differentiable on $\|\boldsymbol{\alpha}\| \leq R_j$ and*

$$\|J_j(\boldsymbol{\alpha})\|_{\mathrm{op}} \leq K_j, \qquad \|J_j(\boldsymbol{\alpha}) - J_j(\boldsymbol{\alpha}')\|_{\mathrm{op}} \leq \Lambda_j \|\boldsymbol{\alpha} - \boldsymbol{\alpha}'\|. \tag{30}$$

On the refiner ball, $\mathcal{A}$ is continuously differentiable and $\|\nabla \mathcal{A}(\boldsymbol{\alpha})\| \leq G_\angle < \infty$. Define

$$\begin{aligned} q_{\mathrm{b}} &= 0, & q_{\mathrm{r}} &= \lambda_\angle G_\angle, \\ \Gamma_j &= K_j(s + K_j R_j) + q_j, & \omega_j &= \Lambda_j R_j (s + K_j R_j) + \tfrac{1}{2} K_j \Lambda_j R_j^2 + q_j. \end{aligned} \tag{31}$$

Each $\mathcal{E}_j(\mathcal{C})$ is a nonempty compact subset of a common compact range $\mathcal{E}_0(\mathcal{C}) \subset (0, \infty)$ fixed before the runs. For a fixed $\zeta_{\mathrm{stab}} \in (0, 2)$, every $\eta \in \mathcal{E}_j(\mathcal{C})$ satisfies

$$\eta \|H_j\|_{\mathrm{op}} \leq 2 - \zeta_{\mathrm{stab}}, \qquad \eta T_j \Gamma_j \leq R_j. \tag{32}$$

The Jacobian bound K_j controls output sensitivity, and Λ_j controls the linearization remainder. The angular-gradient bound contributes q_j to the refiner's update and perturbation bounds. The two step-size conditions keep the tangent recursion stable and both parameter sequences inside the radius R_j through the charged horizon T_j. Compact candidate sets ensure that each optimized risk reduction is attained.

Define the actual and tangent risk reductions by

$$\begin{aligned} \Delta_j(\mathcal{C}, \eta) &= \mathcal{R}_P(0) - \mathcal{R}_P(\boldsymbol{f}_{j,\eta}), & \Delta_j(\mathcal{C}) &= \sup_{\eta \in \mathcal{E}_j(\mathcal{C})} \Delta_j(\mathcal{C}, \eta), \\ \Delta_j^{\mathrm{lin}}(\mathcal{C}, \eta) &= \boldsymbol{b}_j^* H_j^\dagger [I - (I - \eta H_j)^{2T_j}] \boldsymbol{b}_j, & \Delta_j^{\mathrm{lin}}(\mathcal{C}) &= \sup_{\eta \in \mathcal{E}_j(\mathcal{C})} \Delta_j^{\mathrm{lin}}(\mathcal{C}, \eta). \end{aligned} \tag{33}$$

Finally, put

$$\varepsilon_j(\mathcal{C}, \eta) = \tfrac{1}{2} \Lambda_j R_j^2 + K_j \eta T_j \omega_j, \qquad e_j(\mathcal{C}) = \sup_{\eta \in \mathcal{E}_j(\mathcal{C})} 2(s + K_j R_j) \varepsilon_j(\mathcal{C}, \eta). \tag{34}$$

B.2 Finite-budget perturbation and separation

Proposition 6 (Finite-budget tangent certificate)**.** *Under Assumption 5, for every $j \in \{\mathrm{r}, \mathrm{b}\}$ and $\eta \in \mathcal{E}_j(\mathcal{C})$,*

$$\|\boldsymbol{\alpha}_{j,t}^\eta\|, \|\boldsymbol{\beta}_{j,t}^\eta\| \leq t \eta \Gamma_j \leq R_j \quad (0 \leq t \leq T_j), \tag{35}$$

and

$$\|F_j(\boldsymbol{\alpha}_{j,T_j}^\eta) - \Phi_j \boldsymbol{\beta}_{j,T_j}^\eta\| \leq \varepsilon_j(\mathcal{C}, \eta). \tag{36}$$

The tangent reduction is exactly

$$\mathcal{R}_P(0) - \mathcal{R}_P(\Phi_j \boldsymbol{\beta}_{j,T_j}^\eta) = \boldsymbol{b}_j^* H_j^\dagger [I - (I - \eta H_j)^{2T_j}] \boldsymbol{b}_j. \tag{37}$$

Consequently,

$$|\Delta_j(\mathcal{C}, \eta) - \Delta_j^{\mathrm{lin}}(\mathcal{C}, \eta)| \leq 2(s + K_j R_j) \varepsilon_j(\mathcal{C}, \eta), \tag{38}$$

and

$$|\Delta_j(\mathcal{C}) - \Delta_j^{\mathrm{lin}}(\mathcal{C})| \leq e_j(\mathcal{C}). \tag{39}$$

The actual and tangent optimized reductions are attained on their respective candidate sets.

Proof. We derive the tangent reduction and control the perturbation caused by the actual updates through the full charged horizon.

Step 1: Establish the common risk geometry. We first record the common risk geometry. With $\boldsymbol{\eta}_r = \boldsymbol{r} - \bar{\boldsymbol{r}}(W)$, conditional expectation gives $\mathbb{E}[\boldsymbol{\eta}_r \mid W] = 0$. Hence every state-measurable $\boldsymbol{h} \in L^2(P; \mathcal{X})$ satisfies

$$\mathbb{E}\langle \boldsymbol{h}(W), \boldsymbol{\eta}_r \rangle = \mathbb{E}\langle \boldsymbol{h}(W), \mathbb{E}[\boldsymbol{\eta}_r \mid W] \rangle = 0. \tag{40}$$

Expanding the square yields

$$\mathcal{R}_P(\boldsymbol{h}) = \mathbb{E}\|\boldsymbol{r} - \bar{\boldsymbol{r}}\|^2 + \|\bar{\boldsymbol{r}} - \boldsymbol{h}\|^2, \qquad \mathcal{R}_P(0) - \mathcal{R}_P(\boldsymbol{h}) = 2\langle \bar{\boldsymbol{r}}, \boldsymbol{h} \rangle - \|\boldsymbol{h}\|^2. \tag{41}$$

Jensen's inequality also gives $\|\bar{\boldsymbol{r}}\| \leq s$.

Step 2: Keep the actual and tangent iterates in the certified ball. For an in-ball parameter, the line-segment formula and $F_j(0) = 0$ give

$$F_j(\boldsymbol{\alpha}) = \int_0^1 J_j(u\boldsymbol{\alpha})[\boldsymbol{\alpha}], du, \qquad \|F_j(\boldsymbol{\alpha})\| \le K_j\|\boldsymbol{\alpha}\|. \tag{42}$$

Let

$$\boldsymbol{\vartheta}_j(\boldsymbol{\alpha}) = \begin{cases} \lambda_\angle \nabla \mathcal{A}(\boldsymbol{\alpha}), & j = \mathrm{r}, \\ 0, & j = \mathrm{b}. \end{cases} \tag{43}$$

Differentiating equation 41 through F_j gives

$$\nabla L_j(\boldsymbol{\alpha}) = J_j(\boldsymbol{\alpha})^*(F_j(\boldsymbol{\alpha}) - \bar{\boldsymbol{r}}) + \boldsymbol{\vartheta}_j(\boldsymbol{\alpha}). \tag{44}$$

Thus $\|\nabla L_j(\boldsymbol{\alpha})\| \le \Gamma_j$ whenever $\|\boldsymbol{\alpha}\| \le R_j$. Since $\|\Phi_j\|_{\mathrm{op}} \le K_j$, the tangent gradient satisfies

$$\|H_j\boldsymbol{\beta} - \boldsymbol{b}_j\| = \|\Phi_j^*(\Phi_j\boldsymbol{\beta} - \bar{\boldsymbol{r}})\| \le K_j(K_jR_j + s) \le \Gamma_j \tag{45}$$

whenever $\|\boldsymbol{\beta}\| \le R_j$. Starting from zero, simultaneous induction in the two recursions now gives

$$\|\boldsymbol{\alpha}_{j,t+1}^{\eta}\|, \|\boldsymbol{\beta}_{j,t+1}^{\eta}\| \le (t+1)\eta\Gamma_j. \tag{46}$$

The pre-run radius condition makes the right side at most R_j through T_j, completing the induction and proving equation 35.

Step 3: Solve the tangent recursion and compute its risk reduction. We next solve the tangent recursion. The operator $H_j = \Phi_j^*\Phi_j$ is positive semidefinite. Moreover, $\boldsymbol{b}_j \perp \ker H_j$ because $\ker H_j = \ker \Phi_j$. If $H_j\boldsymbol{u}_k = \lambda_k\boldsymbol{u}_k$, the stability certificate gives

$$|1 - \eta\lambda_k| \le 1, \qquad \|(I - \eta H_j)^t\|_{\mathrm{op}} \le 1. \tag{47}$$

The null eigenspaces have zero loading. A finite geometric sum on every positive eigenspace therefore yields

$$\boldsymbol{\beta}_{j,t}^{\eta} = H_j^{\dagger}[I - (I - \eta H_j)^t]\boldsymbol{b}_j. \tag{48}$$

Substitution into equation 41 gives

$$\sum_{k:\lambda_k>0} \frac{|\langle \boldsymbol{u}_k, \boldsymbol{b}_j\rangle|^2}{\lambda_k}[1 - (1 - \eta\lambda_k)^{2T_j}], \tag{49}$$

which is exactly equation 37.

Step 4: Bound the parameter and output perturbations. It remains to transfer the tangent formula to the actual run. The Jacobian Lipschitz condition and the integral formula give

$$\|F_j(\boldsymbol{\alpha}) - \Phi_j\boldsymbol{\alpha}\| \le \frac{1}{2}\Lambda_j\|\boldsymbol{\alpha}\|^2. \tag{50}$$

At the same parameter define $\boldsymbol{\xi}_j(\boldsymbol{\alpha}) = \nabla L_j(\boldsymbol{\alpha}) - (H_j\boldsymbol{\alpha} - \boldsymbol{b}_j)$. Direct subtraction of the two gradients gives

$$\begin{aligned} \boldsymbol{\xi}_j(\boldsymbol{\alpha}) = {} & [J_j(\boldsymbol{\alpha}) - \Phi_j]^*(F_j(\boldsymbol{\alpha}) - \bar{\boldsymbol{r}}) \\ & + \Phi_j^*(F_j(\boldsymbol{\alpha}) - \Phi_j\boldsymbol{\alpha}) + \boldsymbol{\vartheta}_j(\boldsymbol{\alpha}). \end{aligned} \tag{51}$$

For $\|\boldsymbol{\alpha}\| \le R_j$, the three displayed terms are bounded by $\Lambda_jR_j(s + K_jR_j)$, $\frac{1}{2}K_j\Lambda_jR_j^2$ and q_j. Hence

$$\|\boldsymbol{\xi}_j(\boldsymbol{\alpha})\| \le \omega_j. \tag{52}$$

Writing $\boldsymbol{d}_{j,t} = \boldsymbol{\alpha}_{j,t}^{\eta} - \boldsymbol{\beta}_{j,t}^{\eta}$ and subtracting the two recursions gives

$$\boldsymbol{d}_{j,t+1} = (I - \eta H_j)\boldsymbol{d}_{j,t} - \eta\boldsymbol{\xi}_j(\boldsymbol{\alpha}_{j,t}^{\eta}), \qquad \boldsymbol{d}_{j,0} = 0. \tag{53}$$

Equations 35, 47 and 52 imply

$$\|\boldsymbol{d}_{j,t}\| \le \eta\sum_{\ell=0}^{t-1}\|\boldsymbol{\xi}_j(\boldsymbol{\alpha}_{j,\ell}^{\eta})\| \le \eta t\omega_j. \tag{54}$$

Adding and subtracting $\Phi_j\boldsymbol{\alpha}_{j,T_j}^{\eta}$, then using equation 50, proves equation 36.

Step 5: Transfer the output bound to risk reductions. Let $\boldsymbol{f} = \boldsymbol{f}_{j,\eta}$ and $\widetilde{\boldsymbol{f}} = \Phi_j \boldsymbol{\beta}^{\eta}_{j,T_j}$. Both have norm at most $K_j R_j$. The polarization identity and equation 41 give

$$\begin{aligned} |\mathcal{R}_P(\boldsymbol{f}) - \mathcal{R}_P(\widetilde{\boldsymbol{f}})| &\le \|\boldsymbol{f} - \widetilde{\boldsymbol{f}}\|(2\|\bar{\boldsymbol{r}}\| + \|\boldsymbol{f}\| + \|\widetilde{\boldsymbol{f}}\|) \\ &\le 2(s + K_j R_j)\varepsilon_j(\mathcal{C}, \eta), \end{aligned} \tag{55}$$

which proves equation 38. Applying $|\sup u - \sup \boldsymbol{v}| \le \sup |u - \boldsymbol{v}|$ on the same armwise candidate set proves equation 39.

Step 6: Establish attainment on the candidate sets. Finally, the actual gradient is continuous on the certified ball. Finite induction in equation 27 makes the terminal actual output continuous in η. The tangent terminal output is a finite matrix polynomial in η. Both reductions are therefore continuous on the nonempty compact set $\mathcal{E}_j(\mathcal{C})$, so their optima are attained. □

Proof of Theorem 3. We transfer the certified tangent margin to the actual runs, then construct an instance satisfying all the stated conditions.

Step 1: Transfer the tangent advantage to trained risk. Proposition 6 proves the finite-horizon iterate and output bounds, the exact tangent formula and the optimized error certificate. Its two-sided bounds and the strict margin give the positive chain

$$\Delta_{\mathrm{r}} \ge \Delta_{\mathrm{r}}^{\mathrm{lin}} - e_{\mathrm{r}} > \Delta_{\mathrm{b}}^{\mathrm{lin}} + e_{\mathrm{b}} \ge \Delta_{\mathrm{b}}. \tag{56}$$

Both reductions use the common baseline $\mathcal{R}_P(0)$. Subtracting equation 56 from that baseline and using attainment proves equation 14.

Step 2: Construct and bound the angular objective. We finish by constructing the stated nonempty instance. Let $(\zeta_1, \zeta_2) \sim \mathcal{N}(0, I_2)$ and let $\boldsymbol{e} \in \mathcal{X}$ be a unit vector. For $\boldsymbol{\alpha} = (\alpha_1, \alpha_2)$, put

$$s_{\boldsymbol{\alpha}} = \alpha_1\zeta_1 + \alpha_2\zeta_2. \tag{57}$$

With prediction $s_{\boldsymbol{\alpha}}\boldsymbol{e}$, target $\beta\zeta_2\boldsymbol{e}$ and the implemented strict threshold τ, the population angular objective is exactly

$$\mathcal{A}_\beta(\boldsymbol{\alpha}) = 1 - \mathbb{E}[\operatorname{sgn}(s_{\boldsymbol{\alpha}}\zeta_2)\mathbf{1}\{|s_{\boldsymbol{\alpha}}| > \tau, |\zeta_2| > \tau/\beta\}]\,. \tag{58}$$

We verify the gradient envelope consumed below. Let $\varphi(t) = (2\pi)^{-1/2}e^{-t^2/2}$ and $\overline{\Phi}(t) = \int_t^\infty \varphi(s)\,\mathrm{d}s$ denote the standard Gaussian density and upper-tail function. Write $\boldsymbol{\alpha} = \rho(\cos\vartheta, \sin\vartheta)$ and $t = \tau/\rho$. Conditioning the signed integrand on the standard normal $Z_\vartheta = \cos\vartheta\,\zeta_1 + \sin\vartheta\,\zeta_2$, define $m_\vartheta(z)$ as its conditional expectation given $Z_\vartheta = z$. Since $|m_\vartheta| \le 1$, differentiation of

$$\int_{|z|>t} m_\vartheta(z)\varphi(z)\,dz \tag{59}$$

at the moving cutoff gives

$$|\partial_\rho \mathcal{A}_\beta| \le \frac{2\tau}{\rho^2}\varphi(\tau/\rho) \le \frac{4}{e\sqrt{2\pi}\,\tau}. \tag{60}$$

For the angular derivative, use polar Gaussian coordinates

$$(\zeta_1, \zeta_2) = R(\cos\phi, \sin\phi). \tag{61}$$

At each $R > \tau/\rho$, put $\gamma_R = \arccos(\tau/(\rho R))$ and let $q_R(\phi)$ be the signed target mask, which is bounded by one and satisfies $q_R(\phi + \pi) = -q_R(\phi)$. The conditional angular average is

$$\frac{1}{\pi}\int_{\vartheta-\gamma_R}^{\vartheta+\gamma_R} q_R(\phi)\,d\phi. \tag{62}$$

Translating this interval changes the integral at rate at most $2/\pi$. Integration against the polar Gaussian density therefore gives

$$\frac{1}{\rho}|\partial_\vartheta \mathcal{A}_\beta| \le \frac{2}{\pi\rho}e^{-\tau^2/(2\rho^2)} \le \frac{2}{\pi\sqrt{e}\,\tau}. \tag{63}$$

For fixed (ρ, ϑ) with $\rho < \beta$, cutoff intersections occur at at most four radii and have zero polar measure. The uniform translation bound above therefore permits dominated differentiation in R. We also have

$$|\mathcal{A}_\beta(\boldsymbol{\alpha}) - 1| \le 2\overline{\Phi}(\tau/\rho) = o(\rho) \quad \text{as } \rho \downarrow 0, \tag{64}$$

and both displayed derivative bounds converge uniformly to zero. Away from the two cutoff collisions $\boldsymbol{\alpha} = \pm\beta(0,1)$, the same cutoff integrals vary continuously. Thus on every ball $\|\boldsymbol{\alpha}\| \le R < \beta$, the objective is continuously differentiable, $\nabla\mathcal{A}_\beta(0) = 0$ and

$$\|\nabla\mathcal{A}_\beta(\boldsymbol{\alpha})\| \le \left(\frac{4}{e\sqrt{2\pi}} + \frac{2}{\pi\sqrt{e}}\right)\tau^{-1} < \tau^{-1}. \tag{65}$$

Step 3: Realize the correction heads on a sampler state. We now realize these coordinates through a state of the linear-interpolant flow. Draw $\boldsymbol{\xi} \sim \mathcal{N}(0, I_d)$ for $d \ge 2$, take $\mathbf{X}_0 = 0$ and deterministic $\boldsymbol{C}$, let Σ be uniform on $[0,1]$ and set

$$\mathbf{X}_\sigma = \sigma\boldsymbol{\xi}, \qquad \boldsymbol{U} = \boldsymbol{\xi}. \tag{66}$$

Let ζ_1, ζ_2 be two orthonormal coordinates of $\boldsymbol{\xi}$. For $\sigma > 0$, define

$$\boldsymbol{v}_0(\mathbf{X}_\sigma, \sigma, \boldsymbol{C}) = \boldsymbol{\xi} - \beta\zeta_2\boldsymbol{e}. \tag{67}$$

The time law assigns zero mass to $\sigma = 0$, where the checkpoint can be defined arbitrarily. The current state recovers $\boldsymbol{\xi} = \mathbf{X}_\sigma/\sigma$, so the stored feature (ζ_1, ζ_2) is a deterministic function of the current base input. Moreover,

$$\boldsymbol{r} = \bar{\boldsymbol{r}} = \beta\zeta_2\boldsymbol{e}, \qquad s = \beta. \tag{68}$$

Use the actual two-coordinate heads

$$\begin{aligned} F_{\mathrm{r}}(\boldsymbol{\alpha}) &= (\alpha_1\zeta_1 + \alpha_2\zeta_2)\boldsymbol{e}, \\ F_{\mathrm{b}}(\boldsymbol{\alpha}) &= (\alpha_1\zeta_1 + \sqrt{\kappa}\,\alpha_2\zeta_2)\boldsymbol{e}, \qquad 0 < \kappa < 1. \end{aligned} \tag{69}$$

Their Jacobians are constant, so $K_j = 1$ and $\Lambda_j = 0$. Independence and unit variances give

$$H_{\mathrm{r}} = I_2, \quad \boldsymbol{b}_{\mathrm{r}} = (0,\beta)^\top, \qquad H_{\mathrm{b}} = \mathrm{diag}(1,\kappa), \quad \boldsymbol{b}_{\mathrm{b}} = (0,\sqrt{\kappa}\,\beta)^\top. \tag{70}$$

Step 4: Choose admissible parameters and equal resource charges. Set $\tau = 10^{-8}$, $G_\angle = \tau^{-1}$, $Q_\angle = \lambda_\angle G_\angle$, choose $\beta \ge 4Q_\angle$ and then choose

$$\kappa = \frac{1}{4}, \qquad \eta = \frac{1}{4}, \qquad \zeta_{\mathrm{stab}} = 1, \qquad R_{\mathrm{b}} = \frac{\beta}{3}, \qquad R_{\mathrm{r}} = \frac{\beta + Q_\angle}{3}. \tag{71}$$

Equation 65 applies on the refiner ball because $R_{\mathrm{r}} \le 5\beta/12 < \beta$. Choose symmetric two-parameter linear-head schedules with common measured update and state charges c_* and m_*. Let d_* be the added refiner inference charge and declare $\mathcal{C} = (c_*, m_*, d_*)$. Both arms fit this ledger and have horizon one. Fixing the common candidate set to $\{1/4\}$ before either run gives

$$\eta\|H_j\|_{\mathrm{op}} = \frac{1}{4} \le 1, \qquad \eta\Gamma_{\mathrm{b}} = R_{\mathrm{b}}, \qquad \eta\Gamma_{\mathrm{r}} = R_{\mathrm{r}}. \tag{72}$$

Thus every local and resource condition is satisfied.

Step 5: Verify the strict risk-reduction margin. Finally, equation 37 on the loaded eigendirection gives

$$\Delta_{\mathrm{r}}^{\mathrm{lin}} = \beta^2[1 - (1 - 1/4)^2] = \frac{7}{16}\beta^2, \qquad \Delta_{\mathrm{b}}^{\mathrm{lin}} = \beta^2[1 - (1 - 1/16)^2] = \frac{31}{256}\beta^2. \tag{73}$$

Since $\Lambda_j = 0$, we have $\omega_{\mathrm{b}} = 0$, $\omega_{\mathrm{r}} = Q_\angle$, $\varepsilon_{\mathrm{b}} = 0$ and $\varepsilon_{\mathrm{r}} = Q_\angle/4$. Therefore

$$e_{\mathrm{b}} = 0, \qquad e_{\mathrm{r}} = 2(\beta + R_{\mathrm{r}})\frac{Q_\angle}{4} = \frac{Q_\angle(4\beta + Q_\angle)}{6}. \tag{74}$$

The tangent gap is $81\beta^2/256$, while

$$\frac{e_{\mathrm{r}}}{\beta^2} = \frac{1}{6}\frac{Q_\angle}{\beta}\left(4 + \frac{Q_\angle}{\beta}\right) \le \frac{17}{96} < \frac{81}{256}. \tag{75}$$

This proves the strict margin and completes the theorem. □

C Deployment Analysis: Gain, Sampler Law and Endpoint

This section establishes how the correction gain controls deployed velocity error and endpoint distance. We first derive an exact gain quadratic at a fixed state law. We then couple the base and corrected trajectories to certify velocity improvement under each sampler's own law. Finally, we analyze how velocity corrections accumulate along the trajectory and derive an exact endpoint gain identity for affine–Gaussian flows.

C.1 Deployment setting

Fix the trained pair and let $\mathcal{D}_{\mathrm{dep}}$ denote the full deployment domain. Write $\boldsymbol{g}_{\boldsymbol{\phi}}$ for the composite deployed correction $W \mapsto \boldsymbol{g}_{\boldsymbol{\phi}}(\mathcal{Z}(W))$. Fix a measurable reference velocity field

$$\boldsymbol{u}^{\star} : \mathcal{D}_{\mathrm{dep}} \to \mathcal{X}$$

such that

$$\boldsymbol{u}^{\star}(W) = \mathbb{E}[\boldsymbol{U} \mid W] \qquad P\text{-almost surely.}$$

Set

$$\bar{\boldsymbol{r}} = \boldsymbol{u}^{\star} - \boldsymbol{v}_0.$$

For a state law Q and a gain $\kappa \geq 0$, define the deployed velocity error relative to this reference field, together with its alignment and energy coefficients,

$$\mathcal{V}_Q(\kappa) = \mathbb{E}_Q\|\bar{\boldsymbol{r}} - \kappa \boldsymbol{g}_{\boldsymbol{\phi}}\|^2, \qquad A_Q = \mathbb{E}_Q\langle \bar{\boldsymbol{r}}, \boldsymbol{g}_{\boldsymbol{\phi}}\rangle, \qquad B_Q = \mathbb{E}_Q\|\boldsymbol{g}_{\boldsymbol{\phi}}\|^2. \tag{76}$$

On the training law these connect to the risk used above. Taking $\boldsymbol{h} = \kappa \boldsymbol{g}_{\boldsymbol{\phi}}$ in equation 41 gives

$$\mathcal{R}_P(\kappa \boldsymbol{g}_{\boldsymbol{\phi}}) = \mathbb{E}_{\widetilde{P}}\|\boldsymbol{r} - \bar{\boldsymbol{r}}\|^2 + \mathcal{V}_P(\kappa), \tag{77}$$

so $\mathcal{V}_P$ is the entire part of the training risk that the gain moves.

The deployed sampler determines its state law through its trajectory. Running Equation 11 from the prescribed Gaussian source with the composite field $\boldsymbol{v}_0 + \kappa \boldsymbol{g}_{\boldsymbol{\phi}}$, let $\mathbf{X}^{\kappa}(\sigma)$ denote the exact trajectory in the sampler orientation $\sigma : 1 \to 0$ and put

$$Q_{\kappa} = \mathrm{Law}\big(\mathbf{X}^{\kappa}(\Sigma), \Sigma, \boldsymbol{C}\big), \qquad \Sigma \sim q. \tag{78}$$

Here q is a probability law on $[0,1]$, and Σ is independent of the source and condition. The law Q_0 is visited by the uncorrected sampler. We compare these two deployment laws directly in Appendix C.3.

Assumption 6 (Training-law moments)**.** *The realized residual satisfies $\boldsymbol{r} \in L^2(\widetilde{P}; \mathcal{X})$, and the deployed correction satisfies $\boldsymbol{g}_{\boldsymbol{\phi}} \in L^2(P; \mathcal{X})$. The specified field $\bar{\boldsymbol{r}}$ agrees with $\mathbb{E}_{\widetilde{P}}[\boldsymbol{r} \mid W]$ P-almost surely. Conditional Jensen's inequality therefore gives $\bar{\boldsymbol{r}} \in L^2(P; \mathcal{X})$.*

Assumption 7 (Deployment ODE well-posedness)**.** *For a fixed gain radius $\bar{\kappa} > 0$, the family of deployed velocity fields*

$$\{\boldsymbol{v}_0 + \kappa \boldsymbol{g}_{\boldsymbol{\phi}} : \kappa \in [0, \bar{\kappa}]\}$$

is measurable in σ and globally Lipschitz in the state, with a common integrable Lipschitz envelope. Their values at one reference state admit a common integrable envelope over the same gain range. Consequently, their exact ODEs admit unique global solutions on $[0,1]$, and the laws Q_{κ} of equation 78 are well-defined for every $\kappa \in [0, \bar{\kappa}]$.

C.2 Exact gain quadratic and improvement interval

At a fixed state law, the deployed velocity error is an exact quadratic in the gain. Residual–correction alignment A_Q determines the initial slope, and correction energy B_Q determines the curvature. Together, they specify the improvement interval and its optimal gain.

Proposition 7 (Exact gain quadratic)**.** *Let Q be a probability law with $\mathbb{E}_Q\|\bar{\boldsymbol{r}}\|^2 < \infty$ and $\mathbb{E}_Q\|\boldsymbol{g}_{\boldsymbol{\phi}}\|^2 < \infty$. Then, for every $\kappa \in [0, \infty)$,*

$$\mathcal{V}_Q(\kappa) - \mathcal{V}_Q(0) = -2\kappa A_Q + \kappa^2 B_Q. \tag{79}$$

Proof. We verify finiteness of every term before expanding the squared velocity error.

Step 1: Establish integrability on the deployment law. First, A_Q is a finite real number, $B_Q \in [0, \infty)$, $\mathcal{V}_Q(\kappa) < \infty$ for every finite $\kappa \geq 0$ and

$$|A_Q| \leq \mathcal{V}_Q(0)^{1/2} B_Q^{1/2}. \tag{80}$$

By definition $\mathcal{V}_Q(0) = \mathbb{E}_Q\|\bar{\boldsymbol{r}}\|^2 < \infty$ and $B_Q = \mathbb{E}_Q\|\boldsymbol{g}_{\boldsymbol{\phi}}\|^2 \in [0, \infty)$. Cauchy–Schwarz in $L^2(Q; \mathcal{X})$ gives

$$\mathbb{E}_Q|\langle \bar{\boldsymbol{r}}, \boldsymbol{g}_{\boldsymbol{\phi}}\rangle| \leq \big(\mathbb{E}_Q\|\bar{\boldsymbol{r}}\|^2\big)^{1/2}\big(\mathbb{E}_Q\|\boldsymbol{g}_{\boldsymbol{\phi}}\|^2\big)^{1/2} < \infty,$$

which makes A_Q finite and proves equation 80. For a finite gain the pointwise bound $\|\bar{\boldsymbol{r}} - \kappa\boldsymbol{g}_{\boldsymbol{\phi}}\|^2 \leq 2\|\bar{\boldsymbol{r}}\|^2 + 2\kappa^2\|\boldsymbol{g}_{\boldsymbol{\phi}}\|^2$ has an integrable right side, so $\mathcal{V}_Q(\kappa) < \infty$.

Step 2: Expand the gain quadratic at the fixed law. Pointwise in the state, $\|\bar{\boldsymbol{r}} - \kappa\boldsymbol{g}_{\boldsymbol{\phi}}\|^2 = \|\bar{\boldsymbol{r}}\|^2 - 2\kappa\langle \bar{\boldsymbol{r}}, \boldsymbol{g}_{\boldsymbol{\phi}}\rangle + \kappa^2\|\boldsymbol{g}_{\boldsymbol{\phi}}\|^2$. Step 1 makes all three terms integrable and the left side finite, so taking Q-expectations and subtracting $\mathcal{V}_Q(0)$ gives equation 79. Under Q, alignment sets the linear coefficient and correction energy sets the quadratic coefficient. □

Proposition 8 (Sharp gain boundaries)**.** *Under Proposition 7, exactly one of the following holds.*

(i) If $B_Q = 0$, then $\boldsymbol{g}_{\boldsymbol{\phi}} = 0$ Q-almost surely, $A_Q = 0$ and $\mathcal{V}_Q(\kappa) = \mathcal{V}_Q(0)$ for every $\kappa \geq 0$.

(ii) If $B_Q > 0$ and $A_Q \leq 0$, then $\mathcal{V}_Q(\kappa) > \mathcal{V}_Q(0)$ for every $\kappa > 0$.

(iii) If $B_Q > 0$ and $A_Q > 0$, then

$$\mathcal{V}_Q(\kappa) < \mathcal{V}_Q(0) \quad \Longleftrightarrow \quad 0 < \kappa < \frac{2A_Q}{B_Q}, \tag{81}$$

with equality exactly at the two endpoints $\kappa = 0$ and $\kappa = 2A_Q/B_Q$, and $\mathcal{V}_Q(\kappa) > \mathcal{V}_Q(0)$ for every larger gain.

Proof. If $B_Q = 0$, the nonnegative variable $\|\boldsymbol{g}_{\boldsymbol{\phi}}\|^2$ has zero expectation and vanishes Q-almost surely. Then $A_Q = 0$ and $\bar{\boldsymbol{r}} - \kappa\boldsymbol{g}_{\boldsymbol{\phi}} = \bar{\boldsymbol{r}}$ Q-almost surely for every κ, which is case (i).

Suppose $B_Q > 0$. Factoring equation 79 gives

$$\mathcal{V}_Q(\kappa) - \mathcal{V}_Q(0) = \kappa\big(\kappa B_Q - 2A_Q\big). \tag{82}$$

If $A_Q \leq 0$ and $\kappa > 0$, then $\kappa B_Q > 0$ and $-2A_Q \geq 0$, so both factors are positive, which is case (ii). If $A_Q > 0$ and $\kappa > 0$, the sign of equation 82 is the sign of $\kappa B_Q - 2A_Q$, which is negative, zero, or positive according as κ is below, equal to, or above $2A_Q/B_Q$. The separate factor κ supplies equality at $\kappa = 0$. This is case (iii). □

Corollary 9 (Unique optimal gain)**.** *If $A_Q > 0$ and $B_Q > 0$, the unique minimizer of $\kappa \mapsto \mathcal{V}_Q(\kappa)$ on $[0, \infty)$ is*

$$\kappa^\star = \frac{A_Q}{B_Q}, \qquad \mathcal{V}_Q(\kappa^\star) = \mathcal{V}_Q(0) - \frac{A_Q^2}{B_Q} \geq 0. \tag{83}$$

Proof. Completing the square in equation 79 gives

$$\mathcal{V}_Q(\kappa) = \mathcal{V}_Q(0) - \frac{A_Q^2}{B_Q} + B_Q\Big(\kappa - \frac{A_Q}{B_Q}\Big)^2.$$

The last term is nonnegative and vanishes at exactly one scalar, which is positive because $A_Q > 0$ and therefore lies in the allowed domain. Substitution gives the displayed value, and equation 80 gives $A_Q^2/B_Q \leq \mathcal{V}_Q(0)$, which is the stated nonnegativity. □

C.3 Own-law velocity improvement

The next theorem establishes a gain interval over which the corrected sampler achieves lower velocity error than the base sampler, with each field evaluated under its own state law. Coupling the trajectories through the same initial noise and condition lets us control their displacement using spatial regularity. Combining this displacement bound with the fixed-law gain identity gives the comparison between Q_κ and Q_0.

Coupled trajectories. Let $\mathbf{X}_1$ be the prescribed Gaussian source and $\boldsymbol{C}$ the shared condition. For each $\kappa \in [0, \bar{\kappa}]$, use the exact descending ODEs

$$\frac{d\mathbf{X}^0(\sigma)}{d\sigma} = \boldsymbol{v}_0(\mathbf{X}^0(\sigma), \sigma, \boldsymbol{C}), \qquad \mathbf{X}^0(1) = \mathbf{X}_1, \tag{84}$$

$$\frac{d\mathbf{X}^\kappa(\sigma)}{d\sigma} = \boldsymbol{v}_0(\mathbf{X}^\kappa(\sigma), \sigma, \boldsymbol{C}) + \kappa \boldsymbol{g}_{\boldsymbol{\phi}}(\mathbf{X}^\kappa(\sigma), \sigma, \boldsymbol{C}), \qquad \mathbf{X}^\kappa(1) = \mathbf{X}_1. \tag{85}$$

Sample the same independent $\Sigma \sim q$ for both trajectories. Their respective marginals are Q_0 and Q_κ as defined in equation 78. Write $A_0 := A_{Q_0}$ for the alignment along the baseline trajectory.

Assumption 8 (Uniform bounds and spatial regularity)**.** *The bounds below control the magnitude and spatial variation of the residual $\bar{\boldsymbol{r}}$. The reference velocity $\boldsymbol{u}^\star$ determines this field through $\bar{\boldsymbol{r}} = \boldsymbol{u}^\star - \boldsymbol{v}_0$. There is a deterministic common domain that, on an event of probability one, contains the coupled trajectories simultaneously for every $\sigma \in [0, 1]$ and $\kappa \in [0, \bar{\kappa}]$. On this domain, finite nonnegative constants R_r, G, L_r, L_g, L_0 satisfy*

$$\|\bar{\boldsymbol{r}}(\mathbf{x}, \sigma, \boldsymbol{c})\| \le R_r, \qquad \|\boldsymbol{g}_{\boldsymbol{\phi}}(\mathbf{x}, \sigma, \boldsymbol{c})\| \le G, \tag{86}$$

and, for any two states $\mathbf{x}, \mathbf{y}$ in the domain at the same $(\sigma, \boldsymbol{c})$,

$$\|\bar{\boldsymbol{r}}(\mathbf{x}, \sigma, \boldsymbol{c}) - \bar{\boldsymbol{r}}(\mathbf{y}, \sigma, \boldsymbol{c})\| \le L_r \|\mathbf{x} - \mathbf{y}\|, \tag{87}$$

$$\|\boldsymbol{g}_{\boldsymbol{\phi}}(\mathbf{x}, \sigma, \boldsymbol{c}) - \boldsymbol{g}_{\boldsymbol{\phi}}(\mathbf{y}, \sigma, \boldsymbol{c})\| \le L_g \|\mathbf{x} - \mathbf{y}\|, \tag{88}$$

$$\|\boldsymbol{v}_0(\mathbf{x}, \sigma, \boldsymbol{c}) - \boldsymbol{v}_0(\mathbf{y}, \sigma, \boldsymbol{c})\| \le L_0 \|\mathbf{x} - \mathbf{y}\|. \tag{89}$$

The constants are uniform in the noise level, condition, source realization and gain throughout the stated range.

Assumption 7 supplies existence and uniqueness. Assumption 8 quantifies how the correction moves the trajectory and how this displacement affects residual error and alignment across the common deployment domain.

The following theorem gives the full quantitative certificate summarized in the main-text Theorem 4.

Theorem 10 (Detailed own-law velocity certificate)**.** *Under Assumptions 7 and 8, define*

$$H_0(\sigma) := \begin{cases} \dfrac{e^{L_0(1-\sigma)} - 1}{L_0}, & L_0 > 0, \\ 1 - \sigma, & L_0 = 0, \end{cases} \qquad K_Q := G\, \mathbb{E}_{\Sigma \sim q}[H_0(\Sigma)]. \tag{90}$$

Set

$$L_V := 2R_r L_r, \qquad L_A := L_r G + R_r L_g, \qquad M := 2A_0 - L_V K_Q, \qquad N := G^2 + 2L_A K_Q. \tag{91}$$

Then for every $\kappa \in [0, \bar{\kappa}]$,

$$\boxed{\mathcal{V}_{Q_\kappa}(\kappa) - \mathcal{V}_{Q_0}(0) \le -\kappa M + \kappa^2 N.} \tag{92}$$

If $M > 0$, then $N > 0$, and every gain satisfying

$$0 < \kappa \le \bar{\kappa}, \qquad \kappa < \frac{M}{N} \tag{93}$$

gives the strict comparison

$$\boxed{\mathcal{V}_{Q_\kappa}(\kappa) < \mathcal{V}_{Q_0}(0).} \tag{94}$$

Each velocity field is therefore evaluated under the state law generated by its own sampler.

Proof. Fix $\kappa \in [0, \bar{\kappa}]$ and use the common-source coupling throughout.

Step 1: Bound the trajectory displacement. The integral form of equation 85, with the common source cancelled, gives

$$\begin{aligned}\|\mathbf{X}^\kappa(\sigma) - \mathbf{X}^0(\sigma)\| &\le \int_\sigma^1 \|\boldsymbol{v}_0(\mathbf{X}^\kappa(s), s, \boldsymbol{C}) - \boldsymbol{v}_0(\mathbf{X}^0(s), s, \boldsymbol{C})\|\, ds \\ &\quad + \kappa \int_\sigma^1 \|\boldsymbol{g}_{\boldsymbol{\phi}}(\mathbf{X}^\kappa(s), s, \boldsymbol{C})\|\, ds \\ &\le L_0 \int_\sigma^1 \|\mathbf{X}^\kappa(s) - \mathbf{X}^0(s)\|\, ds + \kappa G(1-\sigma). \end{aligned} \tag{95}$$

To handle the descending orientation, put $t = 1 - \sigma$ and $\varphi(t) = \|\mathbf{X}^\kappa(1-t) - \mathbf{X}^0(1-t)\|$. Then

$$\varphi(t) \le L_0 \int_0^t \varphi(s)\, ds + \kappa G t.$$

For $L_0 > 0$, Grönwall's inequality yields

$$\begin{aligned}\varphi(t) &\le \kappa G \left(t + L_0 \int_0^t s e^{L_0(t-s)}\, ds \right) \\ &= \kappa G \frac{e^{L_0 t} - 1}{L_0}. \end{aligned}$$

For $L_0 = 0$, the preceding integral inequality directly gives $\varphi(t) \le \kappa G t$. Thus

$$\|\mathbf{X}^\kappa(\sigma) - \mathbf{X}^0(\sigma)\| \le \kappa G H_0(\sigma), \qquad \mathbb{E}\|\mathbf{X}^\kappa(\Sigma) - \mathbf{X}^0(\Sigma)\| \le \kappa K_Q. \tag{96}$$

The expectation uses the shared source, condition and sampled noise level.

Step 2: Control the change in baseline error. Suppressing the fixed arguments $(\sigma, \boldsymbol{c})$, the residual bounds give

$$\begin{aligned}\big|\|\bar{\boldsymbol{r}}(\mathbf{x})\|^2 - \|\bar{\boldsymbol{r}}(\mathbf{y})\|^2\big| &\le \big(\|\bar{\boldsymbol{r}}(\mathbf{x})\| + \|\bar{\boldsymbol{r}}(\mathbf{y})\|\big)\|\bar{\boldsymbol{r}}(\mathbf{x}) - \bar{\boldsymbol{r}}(\mathbf{y})\| \\ &\le 2R_r L_r \|\mathbf{x} - \mathbf{y}\| = L_V \|\mathbf{x} - \mathbf{y}\|. \end{aligned}$$

Applying this inequality to the coupled states in equation 96 gives

$$\begin{aligned}\mathcal{V}_{Q_\kappa}(0) - \mathcal{V}_{Q_0}(0) &= \mathbb{E}\left[\|\bar{\boldsymbol{r}}(\mathbf{X}^\kappa(\Sigma), \Sigma, \boldsymbol{C})\|^2 - \|\bar{\boldsymbol{r}}(\mathbf{X}^0(\Sigma), \Sigma, \boldsymbol{C})\|^2\right] \\ &\le L_V \mathbb{E}\|\mathbf{X}^\kappa(\Sigma) - \mathbf{X}^0(\Sigma)\| \le \kappa L_V K_Q. \end{aligned} \tag{97}$$

Step 3: Control the change in alignment. At the same fixed $(\sigma, \boldsymbol{c})$,

$$\begin{aligned}&\big|\langle \bar{\boldsymbol{r}}(\mathbf{x}), \boldsymbol{g}_{\boldsymbol{\phi}}(\mathbf{x})\rangle - \langle \bar{\boldsymbol{r}}(\mathbf{y}), \boldsymbol{g}_{\boldsymbol{\phi}}(\mathbf{y})\rangle\big| \\ &\quad \le \big|\langle \bar{\boldsymbol{r}}(\mathbf{x}) - \bar{\boldsymbol{r}}(\mathbf{y}), \boldsymbol{g}_{\boldsymbol{\phi}}(\mathbf{x})\rangle\big| + \big|\langle \bar{\boldsymbol{r}}(\mathbf{y}), \boldsymbol{g}_{\boldsymbol{\phi}}(\mathbf{x}) - \boldsymbol{g}_{\boldsymbol{\phi}}(\mathbf{y})\rangle\big| \\ &\quad \le (L_r G + R_r L_g)\|\mathbf{x} - \mathbf{y}\| = L_A \|\mathbf{x} - \mathbf{y}\|. \end{aligned}$$

Taking expectations under the coupling therefore yields

$$A_{Q_\kappa} \ge A_0 - L_A \mathbb{E}\|\mathbf{X}^\kappa(\Sigma) - \mathbf{X}^0(\Sigma)\| \ge A_0 - \kappa L_A K_Q. \tag{98}$$

Step 4: Combine the fixed-law gain and the trajectory shift. The uniform bounds give $B_{Q_\kappa} \le G^2$ and $\mathcal{V}_{Q_\kappa}(0) \le R_r^2$ for every allowed gain, including zero. Step 1 of the proof of Proposition 7 makes all the risks and coefficients finite. We may therefore apply Proposition 7 at the single law $Q = Q_\kappa$ and then add the change in baseline error:

$$\begin{aligned}&\mathcal{V}_{Q_\kappa}(\kappa) - \mathcal{V}_{Q_0}(0) \\ &\quad = \big[\mathcal{V}_{Q_\kappa}(\kappa) - \mathcal{V}_{Q_\kappa}(0)\big] + \big[\mathcal{V}_{Q_\kappa}(0) - \mathcal{V}_{Q_0}(0)\big] \\ &\quad \le -2\kappa A_{Q_\kappa} + \kappa^2 B_{Q_\kappa} + \kappa L_V K_Q \\ &\quad \le -2\kappa(A_0 - \kappa L_A K_Q) + \kappa^2 G^2 + \kappa L_V K_Q \\ &\quad = -\kappa(2A_0 - L_V K_Q) + \kappa^2(G^2 + 2L_A K_Q) = -\kappa M + \kappa^2 N. \end{aligned} \tag{99}$$

This proves equation 92. If $M > 0$, then $A_0 > 0$. Were $G = 0$, the correction would vanish along the baseline trajectory and force $A_0 = 0$. Hence $G > 0$ and $N \ge G^2 > 0$. For the gains in equation 93, $-\kappa M + \kappa^2 N = \kappa(-M + \kappa N) < 0$, proving equation 94. □

Meaning of the certificate. The constant K_Q converts the correction gain into a bound on expected trajectory displacement. The constants L_V and L_A translate this displacement into bounds on the change in squared baseline residual and residual–correction alignment, respectively. The first-order margin $M = 2A_0 - L_V K_Q$ balances the alignment benefit along the baseline trajectory against the bound on baseline-error change caused by trajectory displacement. The condition $M > 0$ ensures that the alignment benefit exceeds this displacement cost. The second-order coefficient $N = G^2 + 2L_A K_Q$ accounts for correction magnitude and the bound on alignment change along the shifted trajectory. Choosing $\kappa < M/N$ makes the certified first-order benefit dominate these second-order costs. The resulting bound accounts for both the correction and its induced trajectory shift, establishing lower velocity error under the corrected sampler's own state law. Appendix C.4 next analyzes how velocity corrections accumulate into endpoint displacement.

C.4 From velocity error to endpoint distance

The own-law velocity certificate of Theorem 10 does not by itself imply endpoint improvement. Endpoint displacement depends on how velocity errors accumulate over the trajectory. The following proposition identifies the role of temporal cancellation, which motivates the endpoint gain analysis below.

Proposition 11 (Temporal cancellation and endpoint displacement)**.** *There is a one-dimensional instance with state-independent ideal, base and corrected fields, satisfying Assumption 7 on one sampler interval, in which the corrected field has strictly smaller integrated squared velocity error and a strictly smaller one-sided path-length bound than the base field, while its terminal W_2 distance from the ideal law is strictly larger.*

Proof. Reparameterize the sampler interval by $t = 1 - \sigma \in [0, 1]$, so that $t = 0$ is the prescribed source and $t = 1$ is the generated endpoint. This substitution preserves squared velocity error. Let μ be any square-integrable law on $\mathbb{R}$, draw $Z \sim \mu$ and use the same Z as the source of all three flows. Take the ideal drift to be zero, take

$$b(t) = \begin{cases} +1, & 0 \le t < \frac{1}{2}, \\ -1, & \frac{1}{2} \le t \le 1, \end{cases} \tag{100}$$

as the base drift, fix any gain $\kappa_{\mathrm{ce}} \in (0, \bar{\kappa}]$ and define the correction by $g(t) = (\frac{1}{2} - b(t))/\kappa_{\mathrm{ce}}$, so that the corrected drift is the constant $\frac{1}{2}$. All three drifts are measurable, bounded and state independent, hence Lipschitz in the state with constant zero, so they meet Assumption 7.

Direct integration of $\dot{X}^\star = 0$, $\dot{X}^0 = b$ and $\dot{X}^{\mathrm{ce}} = \frac{1}{2}$ from the common source gives

$$X_t^\star = Z, \qquad X_t^0 = \begin{cases} Z + t, & 0 \le t \le \frac{1}{2}, \\ Z + 1 - t, & \frac{1}{2} \le t \le 1, \end{cases} \qquad X_t^{\mathrm{ce}} = Z + \frac{t}{2}, \tag{101}$$

so that $X_1^\star = X_1^0 = Z$ and $X_1^{\mathrm{ce}} = Z + \frac{1}{2}$. The base drift integrates to zero because its two half-interval contributions cancel.

Because both velocity errors are state independent, evaluating them under any common state law gives

$$\int_0^1 \mathbb{E}|0 - b(t)|^2 \, dt = 1, \qquad \int_0^1 \mathbb{E}\Big|0 - \tfrac{1}{2}\Big|^2 dt = \frac{1}{4}, \tag{102}$$

and the one-sided path-length bounds from the common-source coupling are $\int_0^1 |b| \, dt = 1$ and $\int_0^1 \frac{1}{2} \, dt = \frac{1}{2}$. For the endpoints, equation 101 gives $\pi_1^\star = \pi_1^0 = \mu$ while π_1^{ce} is μ translated by $\frac{1}{2}$. The translation coupling gives $W_2(\pi_1^{\mathrm{ce}}, \pi_1^\star) \le \frac{1}{2}$, and for any finite-cost coupling (Y, Y') of the two laws Jensen's inequality gives $\mathbb{E}|Y - Y'|^2 \ge |\mathbb{E}Y - \mathbb{E}Y'|^2 = \frac{1}{4}$. Hence

$$W_2(\pi_1^0, \pi_1^\star) = 0, \qquad W_2(\pi_1^{\mathrm{ce}}, \pi_1^\star) = \frac{1}{2}. \tag{103}$$

These values prove the stated strict inequalities. The signed drift integral determines endpoint displacement in this construction, while squared velocity error measures the drift's energy over time. □

For affine–Gaussian flows with a common linear part, the correction shifts the endpoint mean by an explicit displacement. We use this structure to derive an exact endpoint gain identity. The common linear part propagates the correction through the fundamental solution, and a state-independent correction preserves covariance by translating each conditional law. The square-integrability conditions make the averaged endpoint quadratic finite.

Assumption 9 (Exact affine–Gaussian endpoint specialization)**.** *Conditionally on $\boldsymbol{C} = \boldsymbol{c}$, the ideal and base sampler ODEs are affine Gaussian with a common linear part $M(\sigma)\mathbf{x}$, a common conditional endpoint covariance and an additive drift difference. For $P_{\boldsymbol{C}}$-almost every $\mathbf{c}$, the common linear coefficient and the ideal additive drift are integrable in σ on $[0,1]$, so the ideal ODE has a unique global solution from the prescribed Gaussian source. The deployed correction $\boldsymbol{g}_{\boldsymbol{\phi}}(\sigma, \boldsymbol{c})$ is state independent. Let $\Psi(\sigma, s)$ be the common fundamental solution of $\dot{\mathbf{x}} = M(\sigma)\mathbf{x}$, let $\boldsymbol{m}_0^\star(\boldsymbol{c})$ and $\boldsymbol{m}_0^0(\boldsymbol{c})$ be the ideal and base conditional endpoint means and put $\boldsymbol{E}_0(\boldsymbol{c}) = \boldsymbol{m}_0^\star(\boldsymbol{c}) - \boldsymbol{m}_0^0(\boldsymbol{c})$ and*

$$\boldsymbol{G}(\boldsymbol{c}) = \int_1^0 \Psi(0, s)\, \boldsymbol{g}_{\boldsymbol{\phi}}(s, \boldsymbol{c})\, \mathrm{d}s, \tag{104}$$

in the sampler orientation. Both $\boldsymbol{E}_0$ and $\boldsymbol{G}$ belong to $L^2(P_{\boldsymbol{C}}; \mathcal{X})$.

Proposition 12 (Endpoint displacement and covariance preservation)**.** *Under Assumptions 7 and 9, for $P_{\boldsymbol{C}}$-almost every $\mathbf{c}$ and every finite gain κ in the exact-flow domain, the conditional endpoint law $\pi_0^\kappa(\cdot \mid \boldsymbol{c})$ of the corrected sampler satisfies*

$$\boldsymbol{m}_0^\kappa(\boldsymbol{c}) = \boldsymbol{m}_0^0(\boldsymbol{c}) + \kappa \boldsymbol{G}(\boldsymbol{c}), \qquad \mathrm{Cov}\big(\mathbf{X}_0^\star(\boldsymbol{c})\big) = \mathrm{Cov}\big(\mathbf{X}_0^0(\boldsymbol{c})\big) = \mathrm{Cov}\big(\mathbf{X}_0^\kappa(\boldsymbol{c})\big) = \Xi_0(\boldsymbol{c}). \tag{105}$$

Proof. Fix a condition $\boldsymbol{c}$ and suppress it. Since the correction is state independent, the base and corrected conditional fields are $\mathbf{x} \mapsto M(\sigma)\mathbf{x} + \boldsymbol{b}_0(\sigma)$ and $\mathbf{x} \mapsto M(\sigma)\mathbf{x} + \boldsymbol{b}_0(\sigma) + \kappa \boldsymbol{g}_{\boldsymbol{\phi}}(\sigma, \boldsymbol{c})$, respectively. Let $\mathbf{X}_1$ be the prescribed common Gaussian source. Variation of constants applied directly in the descending orientation gives

$$\mathbf{X}_0^0 = \Psi(0,1)\mathbf{X}_1 + \int_1^0 \Psi(0, s)\boldsymbol{b}_0(s)\, \mathrm{d}s,$$

$$\mathbf{X}_0^\kappa = \Psi(0,1)\mathbf{X}_1 + \int_1^0 \Psi(0, s)\big[\boldsymbol{b}_0(s) + \kappa \boldsymbol{g}_{\boldsymbol{\phi}}(s, \boldsymbol{c})\big]\mathrm{d}s = \mathbf{X}_0^0 + \kappa \boldsymbol{G}(\boldsymbol{c}).$$

Taking conditional expectations gives the first display of equation 105. The same identity is a pathwise translation under the common-source coupling, and $\kappa \boldsymbol{G}(\boldsymbol{c})$ is deterministic once $\boldsymbol{c}$ is fixed, so $\mathbf{X}_0^\kappa - \boldsymbol{m}_0^\kappa(\boldsymbol{c}) = \mathbf{X}_0^0 - \boldsymbol{m}_0^0(\boldsymbol{c})$ and the corrected and base conditional covariances agree. The ideal and base endpoint covariances agree by Assumption 9, which gives the common $\Xi_0(\boldsymbol{c})$.

The terminal-value problem fixes the integration direction from one to zero. For example, if $M \equiv 0$ and $\boldsymbol{g}_{\boldsymbol{\phi}}(s, \boldsymbol{c}) \equiv \boldsymbol{g}$, then equation 104 gives $\boldsymbol{G}(\boldsymbol{c}) = -\boldsymbol{g}$, and the corrected descending flow ends at $\mathbf{X}_0^0 - \kappa \boldsymbol{g}$. □

Proposition 13 (Exact endpoint gain quadratic)**.** *Under Assumption 9 and Proposition 12, put*

$$A_{\mathrm{end}} = \mathbb{E}_{\boldsymbol{C}}\langle \boldsymbol{E}_0, \boldsymbol{G}\rangle, \qquad B_{\mathrm{end}} = \mathbb{E}_{\boldsymbol{C}}\|\boldsymbol{G}\|^2 \ \geq 0. \tag{106}$$

Then for every finite gain κ in the exact-flow domain,

$$\mathbb{E}_{\boldsymbol{C}} W_2^2\big(\pi_0^\kappa, \pi_0^\star\big) - \mathbb{E}_{\boldsymbol{C}} W_2^2\big(\pi_0^0, \pi_0^\star\big) = -2\kappa A_{\mathrm{end}} + \kappa^2 B_{\mathrm{end}}. \tag{107}$$

Proof. We first compute the transport distance for a common covariance, then apply the mean displacement and average over conditions.

Step 1: Compute the Wasserstein distance for a common covariance. For any $\boldsymbol{m}, \boldsymbol{n} \in \mathcal{X}$ and any positive-semidefinite covariance operator Ξ on $\mathcal{X}$,

$$W_2^2\big(\mathcal{N}(\boldsymbol{m}, \Xi), \mathcal{N}(\boldsymbol{n}, \Xi)\big) = \|\boldsymbol{m} - \boldsymbol{n}\|^2. \tag{108}$$

The identity includes singular Ξ and the Dirac case $\Xi = 0$.

Let $\boldsymbol{Z} \sim \mathcal{N}(0, \Xi)$, which is well defined for singular Ξ, and set $\mathbf{X} = \boldsymbol{m} + \boldsymbol{Z}$ and $\mathbf{Y} = \boldsymbol{n} + \boldsymbol{Z}$. This is a coupling of the two laws with $\mathbb{E}\|\mathbf{X} - \mathbf{Y}\|^2 = \|\boldsymbol{m} - \boldsymbol{n}\|^2$, which gives the upper bound. For the matching lower bound, let $(\widetilde{\mathbf{X}}, \widetilde{\mathbf{Y}})$ be any coupling. If its quadratic cost is infinite the bound is immediate, and otherwise Jensen's inequality for $\boldsymbol{z} \mapsto \|\boldsymbol{z}\|^2$ gives $\mathbb{E}\|\widetilde{\mathbf{X}} - \widetilde{\mathbf{Y}}\|^2 \geq \|\mathbb{E}(\widetilde{\mathbf{X}} - \widetilde{\mathbf{Y}})\|^2 = \|\boldsymbol{m} - \boldsymbol{n}\|^2$. Taking the infimum over couplings gives equation 108 for every positive-semidefinite Ξ.

Step 2: Apply the conditional endpoint mean displacement. Proposition 12 gives $\boldsymbol{m}_0^\star(\boldsymbol{c}) - \boldsymbol{m}_0^\kappa(\boldsymbol{c}) = \boldsymbol{E}_0(\boldsymbol{c}) - \kappa \boldsymbol{G}(\boldsymbol{c})$ and a covariance $\Xi_0(\boldsymbol{c})$ shared by all three conditional endpoint laws. Step 1 therefore gives, for $P_{\boldsymbol{C}}$-almost every $\boldsymbol{c}$,

$$W_2^2\big(\pi_0^\kappa(\cdot \mid \boldsymbol{c}), \pi_0^\star(\cdot \mid \boldsymbol{c})\big) = \|\boldsymbol{E}_0(\boldsymbol{c}) - \kappa \boldsymbol{G}(\boldsymbol{c})\|^2, \qquad W_2^2\big(\pi_0^0(\cdot \mid \boldsymbol{c}), \pi_0^\star(\cdot \mid \boldsymbol{c})\big) = \|\boldsymbol{E}_0(\boldsymbol{c})\|^2. \tag{109}$$

These identities express the conditional endpoint distances as squared norms of mean displacements.

Step 3: Average the exact quadratic over conditions. We now average over conditions. Assumption 9 makes $\boldsymbol{E}_0$ and $\boldsymbol{G}$ strongly measurable and square integrable, so both maps in equation 109 are measurable, and $\|\boldsymbol{E}_0 - \kappa \boldsymbol{G}\|^2 \le 2\|\boldsymbol{E}_0\|^2 + 2\kappa^2 \|\boldsymbol{G}\|^2$ makes both integrable. Cauchy–Schwarz gives $\mathbb{E}_{\boldsymbol{C}}|\langle \boldsymbol{E}_0, \boldsymbol{G}\rangle| \le (\mathbb{E}_{\boldsymbol{C}}\|\boldsymbol{E}_0\|^2)^{1/2}(\mathbb{E}_{\boldsymbol{C}}\|\boldsymbol{G}\|^2)^{1/2} < \infty$. Every term is therefore a finite real number, and the pointwise identity $\|\boldsymbol{E}_0 - \kappa \boldsymbol{G}\|^2 - \|\boldsymbol{E}_0\|^2 = -2\kappa\langle \boldsymbol{E}_0, \boldsymbol{G}\rangle + \kappa^2 \|\boldsymbol{G}\|^2$ may be integrated. Substituting equation 109 gives equation 107. □

Corollary 14 (Endpoint gain window)**.** *Under Proposition 13, the corrected sampler has strictly smaller averaged squared endpoint distance than the base sampler exactly when $B_{\mathrm{end}} > 0$, $A_{\mathrm{end}} > 0$ and*

$$0 < \kappa < \frac{2A_{\mathrm{end}}}{B_{\mathrm{end}}}, \tag{110}$$

with equality at both endpoints of that interval and strict degradation beyond it. The unique optimal gain is $A_{\mathrm{end}}/B_{\mathrm{end}}$. If $B_{\mathrm{end}} = 0$, then $\boldsymbol{G} = 0$ almost surely and every gain ties. If $\boldsymbol{E}_0 = 0$ almost surely, then $A_{\mathrm{end}} = 0$ and every positive gain with $B_{\mathrm{end}} > 0$ strictly increases the endpoint distance.

Proof. Equation 107 has the same shape as equation 79 with A_{end} and B_{end} in place of A_Q and B_Q, so Proposition 8 and Corollary 9 apply verbatim to the scalar function $\kappa \mapsto \mathbb{E}_{\boldsymbol{C}} W_2^2(\pi_0^\kappa, \pi_0^\star)$. For the last statement, if $\boldsymbol{E}_0 = 0$ almost surely then $\langle \boldsymbol{E}_0, \boldsymbol{G}\rangle = 0$ almost surely, so $A_{\mathrm{end}} = 0$ and case (ii) of Proposition 8 applies. Step 1 of the proof of Proposition 13 covers every positive-semidefinite conditional covariance Ξ_0, including singular cases. □

D Implementation Details

We expand the settings in Sections 5.1–5.3, covering model configurations, optimization, inference and evaluation.

D.1 Image Generation

Model configuration. We build on the released ImageNet-256 RAEv2 checkpoint (Singh et al., 2026) trained for 80 epochs. The representation autoencoder uses the DINOv3-L encoder with seven-layer feature aggregation and a pretrained ViT-XL decoder. We keep the encoder, decoder and latent normalization statistics fixed throughout adaptation. Each image is represented by a 16×16 grid of 1,024-dimensional latent vectors. The base transformer processes these vectors with latent patch size 1, using 28 blocks of width 1,440 with 20 attention heads, followed by a two-block diffusion head of width 2,048 with 16 attention heads. Its internal guidance branch reads the features after block 8. The refiner uses 10 blocks of width 768, 12 attention heads and an MLP expansion ratio of 4. It takes the predicted clean latent, noise-level and class embeddings and hidden states from base blocks 3, 11, 19 and 27, with block indices counted from one. We initialize the refiner from scratch and zero-initialize its output projection, so its initial velocity correction is zero.

Training. The base retains its flow-matching objective, including supervision of the internal guidance branch. Training noise levels follow the base configuration: we draw a logit-normal variable with underlying Gaussian mean 0 and standard deviation 1, then apply the same factor-8 shift used for sampling below. The refiner minimizes the mean squared error to the realized velocity residual, together with a cosine-direction term of weight 0.1. Its overall loss weight is 1. We detach the residual target and all base-model inputs to the refiner, so the refiner loss updates only the refiner and the base is updated by its own objective. Both networks use GMuon with momentum 0.95, Nesterov updates, zero weight decay and gradient-norm clipping at 1. The base and refiner learning rates warm up from zero over 0.2 epochs to 2×10^{-5} and 4×10^{-4}, respectively, then decay linearly to 2×10^{-6}

and 4×10^{-5} at the end of the 80-epoch adaptation. We train with a global batch size of 1,024 on eight NVIDIA H200 GPUs, using bfloat16 mixed precision, 64 images per GPU per microbatch and two gradient-accumulation steps. The full 80-epoch adaptation comprises 100,080 optimizer steps and takes approximately 42 hours on a single node, including periodic evaluation and checkpointing. Class-conditioning dropout is 0.1, and exponential moving average (EMA) weights are maintained separately for both networks with decay 0.9995.

Sampling and deployment. At test time, we use the EMA weights and generate 50,000 images, with 50 images for each of the 1,000 classes. We start from standard Gaussian latent noise and take 100 Euler steps on the base model's shifted noise schedule. Specifically, a uniform grid $s_i = 1 - i/100$, for $i = 0, \ldots, 100$, is mapped to noise levels $\sigma_i = 8s_i/(1 + 7s_i)$. The shift factor 8 follows the released configuration, $\sqrt{(1024 \cdot 16 \cdot 16)/4096}$. The model predicts clean latents, which the sampler converts to velocities using a denominator $\max(\sigma, 0.05)$ for numerical stability. We evaluate both without guidance and with RAEv2's internal guidance at scale 1.78 over $\sigma \in [0.1, 1]$. In both settings, the refiner correction uses gain $\kappa = 1$ within $\sigma \in [0.15, 0.92]$ and is zero outside this interval. When guidance is enabled, we add the refiner correction to the velocity after the guidance operation. We decode the final latent with the frozen decoder and compute all metrics from the resulting 8-bit RGB images.

Baselines. The published-baseline rows in Table 7 cover SiT-XL/2, DDT-XL, SiT-XL/2-REPA, LightningDiT, REG, REPA-E and RAE-XL. We use their reported evaluations from FD-loss (Yang et al., 2026), whose representation-distance results are also reproduced by RAEv2 (Singh et al., 2026). That evaluation generates 50,000 images from official checkpoints with the corresponding official sampling code, balancing the sample count across classes and using method-specific inference settings. Table 6 summarizes the sampling budgets and guidance mechanisms. The six VAE-based methods use 250 sampling steps with classifier-free guidance (CFG), while RAE-XL uses 50 Euler steps and AutoGuidance from a separately trained weaker model. These published results therefore retain their original sampling budgets and guidance configurations.

Table 6: Evaluation configurations for all seven published image-generation baselines in Table 7. Steps count sampler updates. CFG combines conditional and unconditional predictions, while AutoGuidance combines a main model with a weaker model.

Method	Image representation	Steps	Guidance
SiT-XL/2 (Ma et al., 2024)	SD-VAE	250	CFG
DDT-XL (Wang et al., 2026b)	SD-VAE	250	CFG
SiT-XL/2-REPA (Yu et al., 2025)	SD-VAE	250	CFG
LightningDiT (Yao et al., 2025)	VA-VAE	250	CFG
REG (Wu et al., 2026)	SD-VAE	250	CFG
REPA-E (Leng et al., 2025)	Jointly tuned VAE	250	CFG
RAE-XL (Zheng et al., 2026a)	DINOv2-B RAE	50	AutoGuidance

The RAEv2 models with different training epochs use the same frozen autoencoder, EMA weights, 100-step sampler, class-balanced sample count and guidance configuration as CorrFlow. This comparison isolates the effect of deploying the refiner at a fixed base checkpoint. For SelfFlow (Chefer et al., 2026), we evaluate a local adaptation of its self-supervised flow-matching objective to the same frozen DINOv3-L representation autoencoder. This baseline uses a 30-block transformer of width 1,280 with 20 attention heads, a dual-noise mixing ratio of 0.25 and representation alignment of weight 0.8 between student block 9 and EMA-target block 21. We use EMA weights, the same 100-step shifted Euler grid and 50,000 class-balanced images. The unguided runs disable guidance, and the guided runs use CFG at scale 1.5 over $\sigma \in [0, 0.7]$. The SelfFlow rows report this local RAEv2-latent adaptation rather than the original paper's ImageNet configuration.

Inception-based metrics. We report Inception Score (IS, higher is better) (Salimans et al., 2016) and Fréchet Inception Distance (FID, lower is better) (Heusel et al., 2017). IS measures the confidence and marginal diversity of Inception-v3 class predictions on generated images. For a generated image $\mathbf{x}$, let $p(y \mid \mathbf{x})$ be the classifier distribution over class labels y, and let $p(y)$ be its marginal over generated samples. Then $\mathrm{IS} = \exp\big(\mathbb{E}_{\mathbf{x}} \mathrm{KL}(p(y \mid \mathbf{x}) \| p(y))\big)$, where KL denotes Kullback–Leibler

divergence. Our evaluator shuffles the generated samples before splitting them into ten equal groups and averages the resulting scores. FID compares the mean and covariance of generated and real images in the 2,048-dimensional Inception-v3 feature space. For our local evaluations, both metrics use the Inception-v3-compatible extractor in the RAEv2 evaluation package, with its released ImageNet-256 reference statistics for FID.

Representation-based metrics. We also report Representation Fréchet Distance (FD_r, lower is better) in ConvNeXt, DINOv2 and CLIP feature spaces. These metrics compare generated distributions through complementary supervised, self-supervised and image–text representations. Following the normalized-distance convention of FD-loss (Yang et al., 2026), each table entry is a ratio rather than a raw Fréchet distance. For a frozen feature extractor $\boldsymbol{\phi}$, let $(\boldsymbol{\mu}_G, \Sigma_G)$ and $(\boldsymbol{\mu}_R, \Sigma_R)$ denote the feature mean vector and covariance operator of generated images G and the real training reference R. We compute

$$\mathrm{FD}_{\boldsymbol{\phi}}(G, R) = \|\boldsymbol{\mu}_G - \boldsymbol{\mu}_R\|_2^2 + \mathrm{Tr}\Big(\Sigma_G + \Sigma_R - 2\big(\Sigma_R^{1/2}\Sigma_G\Sigma_R^{1/2}\big)^{1/2}\Big), \tag{111}$$

and normalize it by the distance from real ImageNet validation images V to the same training reference,

$$\mathrm{FD}_{r,\boldsymbol{\phi}}(G) = \frac{\mathrm{FD}_{\boldsymbol{\phi}}(G, R)}{\mathrm{FD}_{\boldsymbol{\phi}}(V, R)}. \tag{112}$$

The ConvNeXt column uses spatially averaged ConvNeXt V2-Base features (Woo et al., 2023) at input resolution 224×224. The DINOv2 and CLIP columns use the class-token features of DINOv2 ViT-L/14 (Oquab et al., 2023) and OpenAI CLIP ViT-L/14 (Radford et al., 2021), respectively, both evaluated at 256×256 resolution. All three feature vectors have dimension 1,024. We retain the released extractor-specific normalization and reference statistics, using bicubic resizing with antialiasing when resizing is needed. The normalization constants in equation 112 are 56.87, 14.19 and 5.60 for ConvNeXt, DINOv2 and CLIP, respectively. A value of 1 corresponds to the real validation-set reference in that feature space. We report the three distances individually to expose differences between representations.

D.2 Text-to-Video Generation

Model configuration. We build on Wan2.1-T2V-1.3B (Wan et al., 2025), keeping its pretrained video variational autoencoder (VAE) and UMT5-XXL text encoder fixed while updating the full base transformer and the refiner. The base transformer has 30 blocks of width 1,536 with 12 attention heads. The VAE maps an 81-frame video at 480×832 resolution to a latent tensor with 16 channels, 21 temporal positions and spatial size 60×104. Both networks use latent patches of size $1 \times 2 \times 2$, giving 32,760 tokens in the same spatiotemporal order. The refiner has 10 blocks of width 1,024, 16 attention heads and an MLP expansion ratio of 4. Its blocks alternate between self-attention with three-dimensional rotary position embeddings and cross-attention to text and base-model features. The base model's noise-level embedding modulates each refiner block through adaptive layer normalization.

We read hidden states after base blocks 4, 12, 20 and 30, corresponding to zero-based code indices 3, 11, 19 and 29. At each token position, we concatenate the four 1,536-dimensional states and fuse them with a $6144 \to 1024 \to 1024$ MLP with a GELU activation. We project the 4,096-dimensional text embeddings to the same width, add learned segment embeddings to distinguish text from hidden-state features and concatenate the two token sequences before layer normalization. The refiner processes the one-step clean prediction and maps its output back to a velocity residual of the original latent shape. It contains 146.58 million parameters, approximately 10.3% of the 1.419-billion-parameter base transformer. The combined transformer and refiner contain 1.566 billion parameters, excluding the frozen VAE and text encoder.

Data and preprocessing. The final training mixture contains 46,572 WISA-80K clips and 20,000 curated OpenVidHD clips, for 66,572 video–text pairs. The WISA subset retains clips with visual-quality scores of at least 4.0 and motion scores of at least 0.1. For WISA, we concatenate the video caption with its physical-law annotation, following the dataset's level-B conditioning. For OpenVidHD (Nan et al., 2025), we use the supplied caption. We shuffle the concatenated datasets without additional source reweighting. Each training example contains 81 frames sampled uniformly across the complete clip, preserving the temporal extent of the depicted event. Short clips reuse

sampled frame indices to obtain the same length. We resize frames while preserving their aspect ratio and center-crop them to 480×832 before VAE encoding.

Training. We jointly train the base model and refiner for five epochs on the processed 66,572-clip dataset using AdamW, with learning rates of 5×10^{-6} and 10^{-4}, respectively, momentum coefficients $(0.9, 0.999)$ and weight decay 0.01. The learning rates start at one-third of these values for five scheduler steps and then remain at the nominal rates. Training uses four NVIDIA H200 GPUs with a global batch size of 16, bfloat16 mixed precision and gradient checkpointing, taking approximately 201 hours including checkpointing. The base uses the scheduler-weighted flow-matching loss, while the refiner minimizes residual mean squared error plus a cosine-direction loss weighted by 0.1, with unit overall refiner-loss weight. All three benchmarks use the resulting paired weights.

Sampling and deployment. We generate 81-frame videos at 480×832 resolution and save them at 16 frames per second. Sampling starts from standard Gaussian latent noise and uses 50 Euler updates with noise shift 5 and classifier-free guidance (CFG) scale 5. For $s_i = 1 - i/50$, with $i = 0, \ldots, 50$, the noise grid is $\sigma_i = 5s_i/(1 + 4s_i)$. We use the fixed Wan negative prompt and apply the refiner to the positive text branch with gain $\kappa = 0.2$ when $\sigma \in [0.15, 0.92]$. Let $\boldsymbol{v}_+$ and $\boldsymbol{v}_-$ denote the base velocities conditioned on the benchmark prompt and negative prompt, respectively, and let $\hat{\boldsymbol{r}}_+$ denote the positive-branch residual prediction. At active steps, the guided velocity is

$$\boldsymbol{v}_{\text{sample}} = \boldsymbol{v}_- + 5\big(\boldsymbol{v}_+ + 0.2\hat{\boldsymbol{r}}_+ - \boldsymbol{v}_-\big) = \boldsymbol{v}_- + 5(\boldsymbol{v}_+ - \boldsymbol{v}_-) + \hat{\boldsymbol{r}}_+. \tag{113}$$

Thus the branch-level gain yields a unit-strength residual contribution to the guided velocity. We recompute the clean prediction and hidden states from the current latent at every active step. The base runs in bfloat16 and the refiner in float32, and the frozen VAE decodes the final latent to RGB frames.

VBench. We use the released VBench prompt suite and evaluators (Huang et al., 2024) to measure 16 dimensions of video quality and text alignment. Seven dimensions measure visual quality and temporal behavior, and nine measure semantic consistency. Tables 8 and 9 display dimension scores on a 0–100 scale. Quality Score and Semantic Score use the benchmark's dimension-specific normalization and weights, and Overall Score combines them with weights 0.8 and 0.2, respectively. Temporal Flickering measures resistance to frame-to-frame flicker, so a larger score indicates better temporal stability. Dynamic Degree measures the presence of motion through optical flow and complements Motion Smoothness, which evaluates temporal regularity.

VideoPhy2. We evaluate the 590-prompt test set, including its 180-prompt HARD subset, with the released VideoPhy2 automatic evaluator (Bansal et al., 2026). Generation uses the benchmark's expanded captions, while scoring uses the original action captions. The evaluator assigns ratings from 1 to 5 for physical commonsense (PC) and semantic adherence (SA). The PC and SA columns report the percentage of ratings at least 4, and Joint reports the percentage satisfying both thresholds. Joint therefore measures whether the requested event is depicted with plausible physics, combining two requirements that the separate scores assess individually.

FinePhyEval. We evaluate all 65 FinePhyEval prompts using their released question graphs (Pothiraj et al., 2026). These graphs decompose each event into object existence, action execution and physical behavior. Our local evaluation uses GPT-5.5 to answer the fixed questions from sampled video frames in the two-step PQSG question-answering procedure. A failed prerequisite propagates failure to dependent questions, so a physics judgment also requires the relevant objects and actions to be present. Object, Action and Physics are the fractions of satisfied questions in each category after this propagation, with all questions retained in their respective denominators. Overall is the corresponding fraction across all questions, weighting categories by their question counts. All four scores lie between 0 and 1.

Comparison methods. We compare with the released Wan2.1 models and methods that adapt video generators for motion or physical consistency, including VideoJAM (Chefer et al., 2025), DreamWorld (Tan et al., 2026), VPT (Zheng et al., 2026b), VideoREPA (Zhang et al., 2025c) and PhysRAG (Cheng et al., 2026), as available for each benchmark. The model-size suffixes identify the base-generator family, while the refiner parameter count is reported above. The tables combine published benchmark results with local evaluations of released implementations, retaining method-specific generation settings.

D.3 IMAGE-TO-VIDEO GENERATION

Model configuration. We build on Wan2.2-TI2V-5B (Wan et al., 2025), whose base transformer has 30 blocks of width 3,072 with 24 attention heads. As in Appendix D.2, we freeze the video VAE and text encoder and update the full base transformer together with the refiner. The Wan2.2 VAE encodes an 81-frame training video at 704×1248 resolution into 48 latent channels with 21 temporal positions and spatial size 30×52. The refiner uses 13 blocks of width 2,048 with 32 attention heads. Its latent patch size, MLP expansion ratio, alternating attention blocks, hidden-state selection and conditioning design are the same as in text-to-video generation. To match the larger base features, its fusion MLP maps the concatenated hidden states through $12288 \to 2048 \to 2048$, and its noise-level projection accepts 3,072-dimensional embeddings. The refiner predicts a 48-channel velocity residual and is initialized from scratch with a zero-initialized output projection.

Training. The training mixture, caption construction, frame sampling and spatial preprocessing are the same as in Appendix D.2. We also use the same five-epoch schedule, global batch size, AdamW configuration, learning rates, loss weights, mixed precision and gradient checkpointing. For each training example, we enable image conditioning with probability 0.4 by encoding the first sampled frame with the frozen VAE. For these examples, we replace the first temporal latent with this clean image latent and assign its tokens noise level zero in the base transformer. Both the base flow-matching loss and the refiner residual loss are computed over the remaining temporal latents. The other examples use text conditioning alone. We detach the residual target and base-model inputs to the refiner, so each network is updated by its own objective. Training begins on 16 NVIDIA H200 GPUs with one clip per GPU and resumes after 4,200 optimizer steps on four H200 GPUs with four gradient-accumulation steps, preserving the global batch size of 16. We evaluate the paired base and refiner weights at the end of the fifth epoch.

Sampling and image conditioning. We generate at 704×1248 resolution, using the same video length, frame rate, Euler sampler, noise grid, CFG scale, negative prompt, refiner gain and active noise interval as in Appendix D.2. The positive-branch correction follows equation 113, and the base and refiner use the same inference precisions as in text-to-video generation. For each benchmark example, we supply the released image at the start of the continuation and its corresponding text prompt, including the benchmark's fixed-camera instructions and action description. We resize the image to the generation resolution, encode it with the frozen VAE and keep its latent fixed at the first temporal position. We restore this condition in the refiner's clean prediction and after every sampling update.

Physics-IQ Verified. The image-to-video evaluation covers 66 physical scenarios from three camera viewpoints, yielding 198 image–text pairs. We retain the first 80 frames of each generated video for the benchmark's five-second evaluation at 16 frames per second. The released evaluator compares generated continuations with recorded reference videos at 540×960 resolution, using motion masks and RGB frames without an LLM judge. Spatial measures the intersection-over-union of regions that exhibit motion anywhere in the clip. Spatio-temporal measures motion-mask overlap at corresponding frames, while Weighted Spatial compares how frequently each pixel belongs to a moving region. MSE measures the mean squared difference between generated and reference RGB frames.

For each scenario and viewpoint, the evaluator normalizes these measurements by the corresponding comparison between two real recordings of the same event. The three overlap metrics are divided by this real-to-real reference, while MSE uses the inverse ratio so that higher is better. Each normalized component is clipped to $[0, 1]$, and Verified Score averages the four components and then the scenario–viewpoint pairs. Table 4 reports all scores on a 0–100 scale. All rows in Table 4 use the benchmark's setting without LLM support.

Yuan et al. (2026) describe two inference-time uses of the WMReward signal: gradient guidance, which steers each denoising trajectory with the reward gradient, and Best-of-N selection, which draws multiple guided samples and keeps the one with the highest reward. We enable only reward guidance and generate a single sample per condition, without Best-of-N selection. Because Yuan et al. (2026) attribute most of their Physics-IQ improvement to the selection step, the WMReward row is expected to score below their guidance-plus-selection results. This choice keeps the comparison at a matched sampling budget. The other baselines are run using their default configurations as released.

E More Results

E.1 More Results on ImageNet Generation

E.1.1 Quantitative Results

Table 7 extends the comparison in Table 1 with additional methods, training budgets, base-only evaluations and FID scores.

Table 7: Comparison on ImageNet-256 (Deng et al., 2009) generation. For published baselines, the FID column reports guided FID (gFID) as in the original papers. Models are evaluated with and without guidance as indicated. For our method, $80 + 40$ and $80 + 80$ indicate that we start from the 80-epoch RAEv2 checkpoint and train our refiner for an additional 40 and 80 epochs, respectively. "Ours (base only)" evaluates the corresponding base model without the refiner at inference and reports the total base-model training epochs. Representation Fréchet Distance (FD_r) is reported in the feature spaces of ConvNeXt (Liu et al., 2022), DINOv2 (Oquab et al., 2023) and CLIP (Radford et al., 2021).

Method	Epochs	Guidance	IS ↑	FID ↓	Representation Fréchet Distance (FD_r) ↓		
					ConvNeXt	DINOv2	CLIP
Published baselines							
SiT-XL/2 (Ma et al., 2024)	800	Yes	256.7	2.12	2.02	7.89	17.69
DDT-XL (Wang et al., 2026b)	800	Yes	**309.3**	1.26	1.02	4.26	13.86
SiT-XL/2-REPA (Yu et al., 2025)	800	Yes	306.1	1.42	1.22	4.27	12.65
LightningDiT (Yao et al., 2025)	800	Yes	294.3	1.42	1.09	3.76	10.21
REG (Wu et al., 2026)	800	Yes	302.9	1.54	1.14	3.45	10.86
REPA-E (Leng et al., 2025)	800	Yes	298.3	1.17	1.28	2.44	6.28
RAE-XL (Zheng et al., 2026a)	800	Yes	261.0	1.16	1.79	2.11	7.87
Evaluations without guidance							
RAEv2 (Singh et al., 2026)	80	No	225.21	1.6038	1.4127	2.3965	7.3809
RAEv2 (Singh et al., 2026)	120	No	230.53	1.5317	1.3050	2.2879	7.2028
SelfFlow (Chefer et al., 2026)	120	No	144.58	4.8232	4.8635	7.1880	12.2924
Ours	80+40	No	231.96	1.5582	1.2818	2.2635	7.2213
RAEv2 (Singh et al., 2026)	160	No	233.94	1.5070	1.2753	2.2226	7.1804
SelfFlow (Chefer et al., 2026)	160	No	146.37	4.7480	4.8110	7.0750	12.1329
Ours	80+80	No	234.02	1.5367	1.2396	2.2051	7.1445
Evaluations with guidance							
RAEv2 (Singh et al., 2026)	80	Yes	260.44	1.0567	0.7745	1.1020	5.2134
RAEv2 (Singh et al., 2026)	120	Yes	265.54	**1.0235**	0.7451	1.0732	4.8667
SelfFlow (Chefer et al., 2026)	120	Yes	145.03	4.7948	4.8228	7.0984	12.2674
Ours	80+40	Yes	264.95	1.0446	0.7475	1.0869	4.8648
RAEv2 (Singh et al., 2026)	160	Yes	264.97	1.0337	0.7363	1.0797	4.8549
SelfFlow (Chefer et al., 2026)	160	Yes	146.79	4.6462	4.7155	6.9849	12.1602
Ours	80+80	Yes	266.60	1.0593	**0.7258**	**1.0339**	**4.8415**

E.1.2 Qualitative Results

Figures 5 and 6 compare selected samples from Self-Flow, RAEv2 and CorrFlow on class-conditional ImageNet generation at 256×256 resolution across eight categories. In these examples, CorrFlow produces clearer object structures and more recognizable category-specific details, including the lattice of shoji screens, the heads and handles of hatchets and the gills of axolotls.

At the same 160-epoch base-model training budget, CorrFlow improves all three FD_r metrics over RAEv2 both with and without guidance, consistent with the clearer shoji lattices, hatchet geometry and axolotl gills in Figures 5 and 6. With guidance, IS increases from 264.97 to 266.60 and DINOv2 FD_r decreases from 1.0797 to 1.0339, although FID worsens from 1.0337 to 1.0593 (Table 7). The base model exhibits a similar discrepancy: continuing RAEv2 fine-tuning from 120 to 160 total epochs raises guided FID from 1.0235 to 1.0337, while ConvNeXt and CLIP FD_r continue to decrease. These discrepancies suggest practical saturation of FID as a perceptual ranking signal in this regime near 1, where small score differences are an unreliable standalone indicator of generation quality. FID compares only the means and covariances of Inception-v3 features, so improvements in object structure and fine detail need not yield a lower score. Jayasumana et al. (2024) document disagreements between FID and human judgments and show that FID can fail to track progressive visual improvements, while Singh et al. (2026) observe that incremental gFID gains provide little

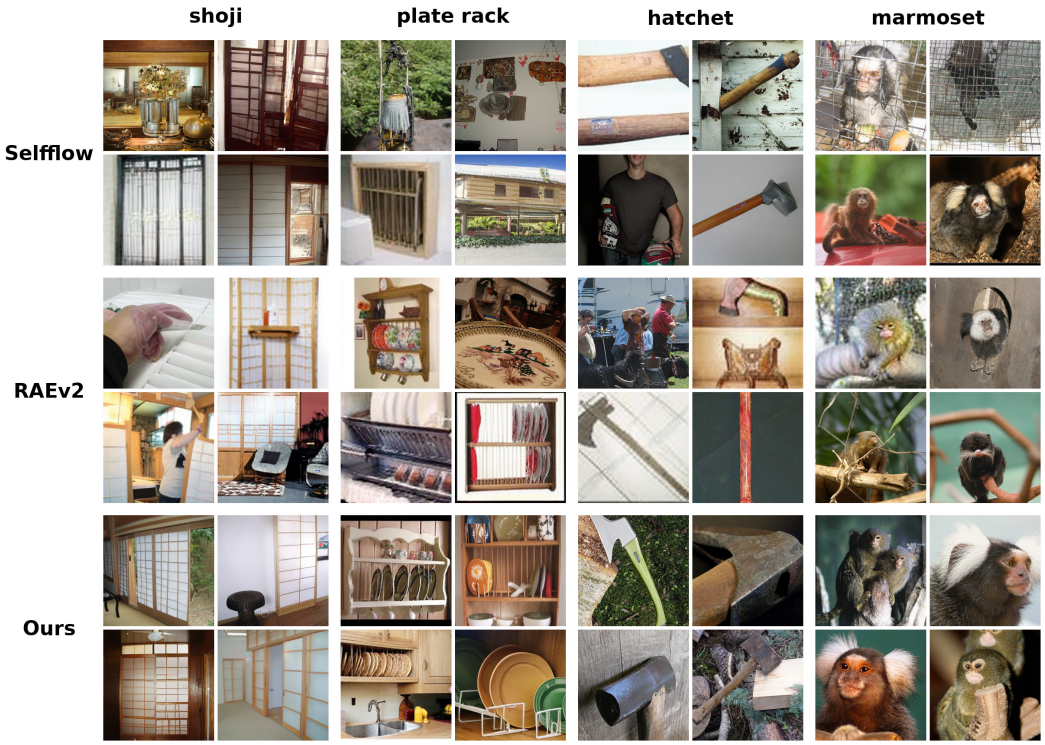

Figure 5: Qualitative results on ImageNet at 256×256 resolution. Columns show shoji, plate rack, hatchet and marmoset. Row groups show Self-Flow (Selfflow), RAEv2 and CorrFlow (Ours). Each method–category block contains four samples in a 2×2 grid, giving 48 images in total. All three base DiTs are trained for 160 epochs. The CorrFlow refiner is trained for 80 epochs.

practical signal for strong RAEv2 models. The consistent improvements across complementary representation spaces, together with the visual examples, support the benefit of learned residual corrections beyond what FID alone captures.

E.2 More Results on Text-to-Video Generation

E.2.1 Quantitative Results

Tables 8 and 9 report the full VBench results for the methods in Table 3, covering all 16 evaluation dimensions and the quality, semantic and overall scores.

Table 8: VBench results (Part 1/2): consistency, motion and object-level metrics. All scores are reported on a 0–100 scale.

Method	Subject Consistency	Background Consistency	Temporal Flickering	Motion Smoothness	Dynamic Degree	Aesthetic Quality	Imaging Quality	Object Class	Multiple Objects	Human Action
VideoREPA-5B (Zhang et al., 2025c)	95.47	95.10	98.23	98.54	47.22	57.19	59.80	67.56	40.32	**83.00**
PhysRAG-5B (Cheng et al., 2026)	96.31	95.66	98.71	96.93	**81.94**	56.80	65.56	80.50	**75.00**	63.00
Wan 2.1-1.3B (Wan et al., 2025)	91.83	94.71	99.17	96.51	65.00	57.24	59.14	76.09	53.66	74.60
VideoJAM-1.3B (Chefer et al., 2025)	91.51	96.01	99.13	96.05	73.88	58.41	62.68	79.22	59.66	79.20
DreamWorld-1.3B (Tan et al., 2026)	93.62	94.95	98.81	98.07	79.16	59.59	66.76	81.32	65.03	81.20
VPT-1.3B (Zheng et al., 2026b)	92.65	**97.27**	98.64	97.85	70.83	58.09	63.68	73.43	51.92	74.80
CorrFlow (Ours)-1.3B	**96.68**	96.95	**99.81**	**98.85**	63.54	**61.65**	**68.66**	**82.81**	67.72	78.00

Visual quality and temporal consistency. CorrFlow achieves the highest VBench Quality Score (84.69) and Overall Score (81.86) among the compared methods. Relative to Wan2.1-T2V-1.3B, these gains are 4.88 and 4.93 points, respectively. The improvements span both individual-frame quality and consistency over time. Imaging Quality increases from 59.14 to 68.66, Aesthetic Quality from 57.24 to 61.65 and Subject Consistency from 91.83 to 96.68. Temporal Flickering and Motion

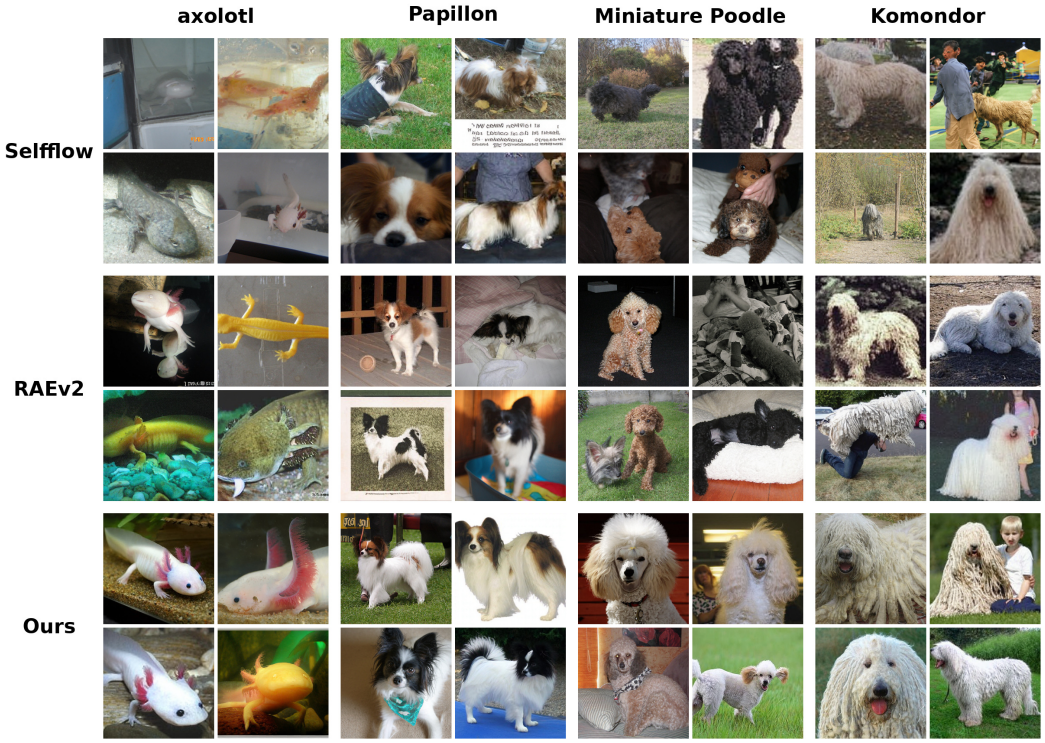

Figure 6: Qualitative results on ImageNet at 256 × 256 resolution. Columns show axolotl, Papillon, Miniature Poodle and Komondor. Row groups show Self-Flow (Selfflow), RAEv2 and CORRFLOW (Ours). Each method–category block contains four samples in a 2 × 2 grid, giving 48 images in total. All three base DiTs are trained for 160 epochs. The CORRFLOW refiner is trained for 80 epochs.

Table 9: VBench results (Part 2/2): appearance, style and overall scores. All scores are reported on a 0–100 scale.

Method	Color	Spatial Relationship	Scene	Appearance Style	Temporal Style	Overall Consistency	Quality Score	Semantic Score	Overall Score
VideoREPA-5B (Zhang et al., 2025c)	86.12	51.85	**32.27**	21.37	22.71	24.20	79.96	63.53	76.67
PhysRAG-5B (Cheng et al., 2026)	**93.95**	77.33	15.55	22.00	23.18	23.18	83.07	68.12	80.08
Wan 2.1-1.3B (Wan et al., 2025)	89.93	62.37	20.03	**22.91**	22.23	24.25	79.81	65.43	76.93
VideoJAM-1.3B (Chefer et al., 2025)	88.92	66.17	28.34	22.67	24.03	**25.01**	81.18	69.08	78.76
DreamWorld-1.3B (Tan et al., 2026)	92.61	70.47	29.71	22.14	23.82	24.87	83.49	**70.89**	80.97
VPT-1.3B (Zheng et al., 2026b)	86.99	73.35	20.13	20.87	22.52	23.42	82.26	64.91	78.79
CORRFLOW (Ours)-1.3B	91.71	**77.88**	25.15	20.98	**24.05**	24.31	**84.69**	70.55	**81.86**

Smoothness also improve, reaching 99.81 and 98.85. Together, these metrics indicate that the adapted model improves the appearance of generated frames while maintaining more coherent subjects and smoother temporal evolution. Compared with DreamWorld, the strongest alternative by Overall Score in this table, CORRFLOW improves Quality Score by 1.20 points and Overall Score by 0.89 points.

Semantic alignment and composition. CORRFLOW improves prompt alignment across object identity, composition and action. Relative to Wan2.1-T2V-1.3B, Object Class rises from 76.09 to 82.81, Multiple Objects from 53.66 to 67.72 and Spatial Relationship from 62.37 to 77.88. These improvements support more accurate rendering of the requested entities and their arrangement, and Semantic Score increases from 65.43 to 70.55. Human Action increases from 74.60 to 78.00, and Temporal Style from 22.23 to 24.05. The gains therefore extend from the identity and spatial arrangement of objects to the actions and temporal style specified by the prompt.

Physical plausibility with semantic adherence. On VideoPhy2 (Table 2), CORRFLOW improves the ALL-set PC score from 57.8 to 65.7 and SA from 30.0 to 31.3 relative to Wan2.1-T2V-1.3B. Joint increases by 2.2 percentage points to 27.0, the highest score in the comparison. This Joint score

also exceeds those of Wan2.1-14B (24.7) and VideoREPA (23.6). On HARD, CORRFLOW raises PC from 36.7 to 51.9 and Joint from 5.6 to 6.9, corresponding to gains of 15.2 and 1.3 percentage points. The improvements on both test splits show that more generated videos depict the requested event while satisfying physical commonsense.

Object, action and physics requirements. FinePhyEval provides a complementary decomposition of these gains. Relative to Wan2.1-T2V-1.3B, CORRFLOW improves Object from 0.84 to 0.90, Action from 0.49 to 0.55, Physics from 0.36 to 0.44 and Overall from 0.54 to 0.60. It ties DreamWorld on Object and obtains the highest Action and Overall scores among the compared methods. Its Overall score of 0.60 also exceeds the 0.59 achieved by Wan2.1-14B. The question-graph metric rewards physical behavior together with the objects and actions on which it depends. The gains across all three categories therefore support more complete realization of the requested physical events, complementing the visual and temporal improvements measured by VBench.

E.2.2 QUALITATIVE RESULTS

Figures 7, 8 and 9 provide additional qualitative comparisons of VideoREPA, VPT, PhysRAG and CORRFLOW (Ours). The five examples cover domino interactions, multiple-object composition, canoe paddling, jetski racing and parasailing. For each case, rows follow this method order from top to bottom, and columns show five frames sampled at evenly spaced relative positions across each video, progressing from left to right. The text below each case gives the corresponding prompt description.

E.3 MORE RESULTS ON IMAGE-TO-VIDEO GENERATION

E.3.1 QUANTITATIVE RESULTS

Aggregate physical consistency. CORRFLOW achieves the highest Verified Score in Table 4, reaching 30.95 compared with 27.71 for Wan2.2-5B. The gain is 3.24 points over the base model and 4.47, 3.52 and 5.41 points over WMReward, VLIPP and PhysRAG, respectively. This score combines motion-region agreement, temporal overlap and frame reconstruction after normalization by variation between real recordings.

Motion coverage and frame appearance. Relative to Wan2.2-5B, CORRFLOW increases Spatial from 35.89 to 43.43 and Weighted Spatial from 22.96 to 24.74. These gains indicate closer agreement with the reference in both the regions traversed by moving objects and the fraction of time each region exhibits motion. MSE Score increases from 25.54 to 29.24, indicating improved agreement in RGB frame appearance under the benchmark's normalization. CORRFLOW leads the compared methods on all three metrics, supporting the aggregate improvement across complementary aspects of the generated continuation.

E.3.2 QUALITATIVE RESULTS

Figure 4 compares CORRFLOW with WMReward and PhysRAG under the same image and text conditions. In the moving-light example, CORRFLOW retains a recognizable statuette silhouette across the displayed frames as its shadow changes position. The mandarin example shows the knife and fruit remaining identifiable through the slicing motion. These examples illustrate object and shadow consistency alongside the motion and appearance metrics above.

Figures 10 and 11 extend this comparison to three additional scenarios, with ground-truth videos (GT) alongside WAN, VLIPP, WMReward, PhysRAG and CORRFLOW. The paper-in-water example shows how each method depicts the paper after release and preserves the surrounding scene, while the mirror examples show the relationship between object motion and its reflection. Each row presents five video frames in temporal order.

Figure 12 adds a bouncing tennis ball and a blue ball rolling from a table onto glass, with the same six rows per case. These examples show how each model continues object motion from the initial frame. Red boxes mark the visible balls and adjacent reflections in the four baseline methods, while GT and CORRFLOW remain unmarked.

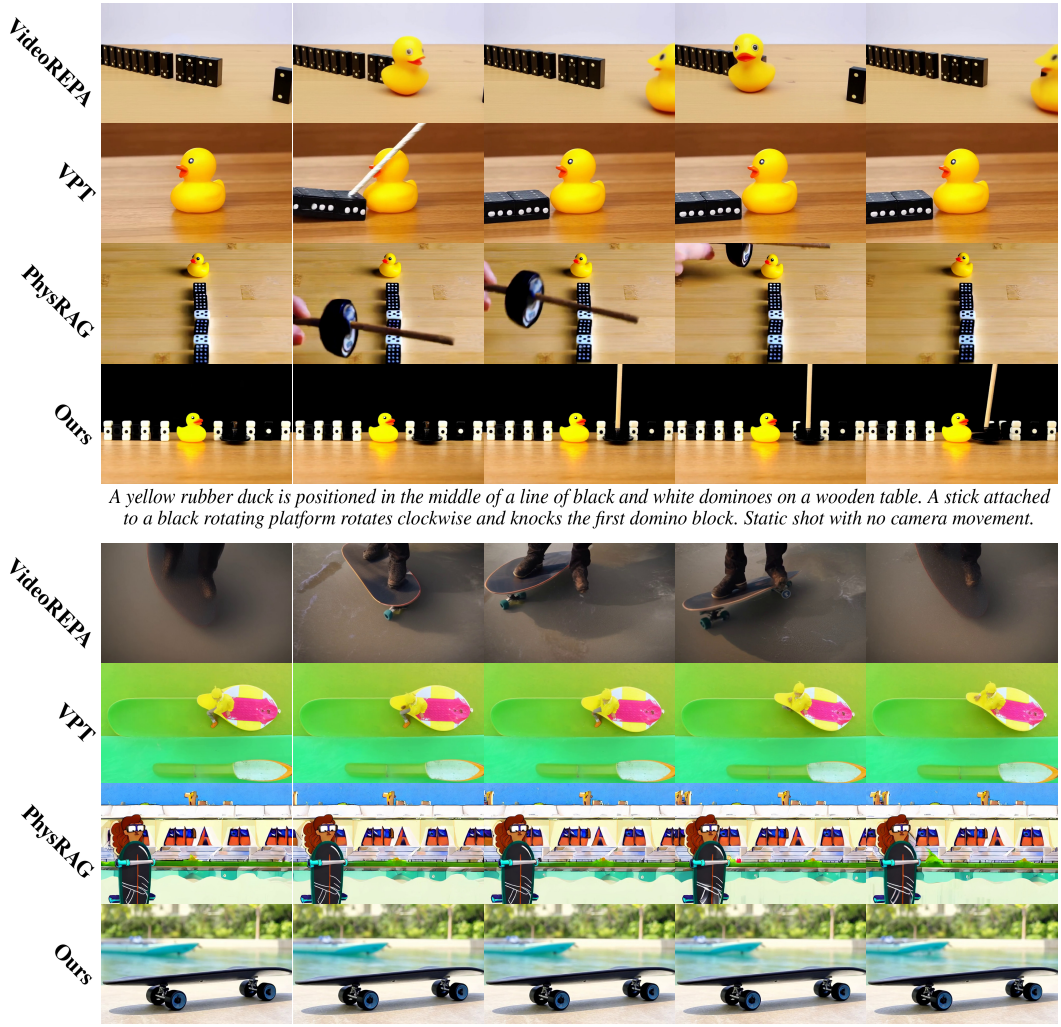

Figure 7: **Qualitative results on text-to-video generation.** The upper case shows a rubber duck among dominoes, and the lower case shows a skateboard and a surfboard. Within each case, rows show VideoREPA, VPT, PhysRAG and Ours from top to bottom. Each row contains five frames sampled at evenly spaced relative positions across the video, ordered from left to right.

E.4 Full-Precision Ablation Results

Table 10 reports all ablations from Section 5.4 on the original 0–1 scale with four decimal places. Table 5 displays the same scores as percentages with three significant digits, while retaining the ranking colors determined at four-decimal precision. Rounding can make distinct scores appear tied. For example, the Imaging Quality scores of 0.6369 and 0.6368 both display as 63.7%, and the Quality Scores of 0.7980 and 0.7979 both display as 79.8%.

F Reproducibility Statement

To support reproducibility, Section 3 specifies the refiner architecture, residual training objective and inference procedure of CorrFlow. Appendix D documents the pretrained models, training data, preprocessing, optimization hyperparameters, hardware and sampling settings for image, text-to-video and image-to-video generation, together with the benchmark protocols and metric definitions. Section 5.4 describes the ablation setup, and Appendix E.4 provides the corresponding scores at full

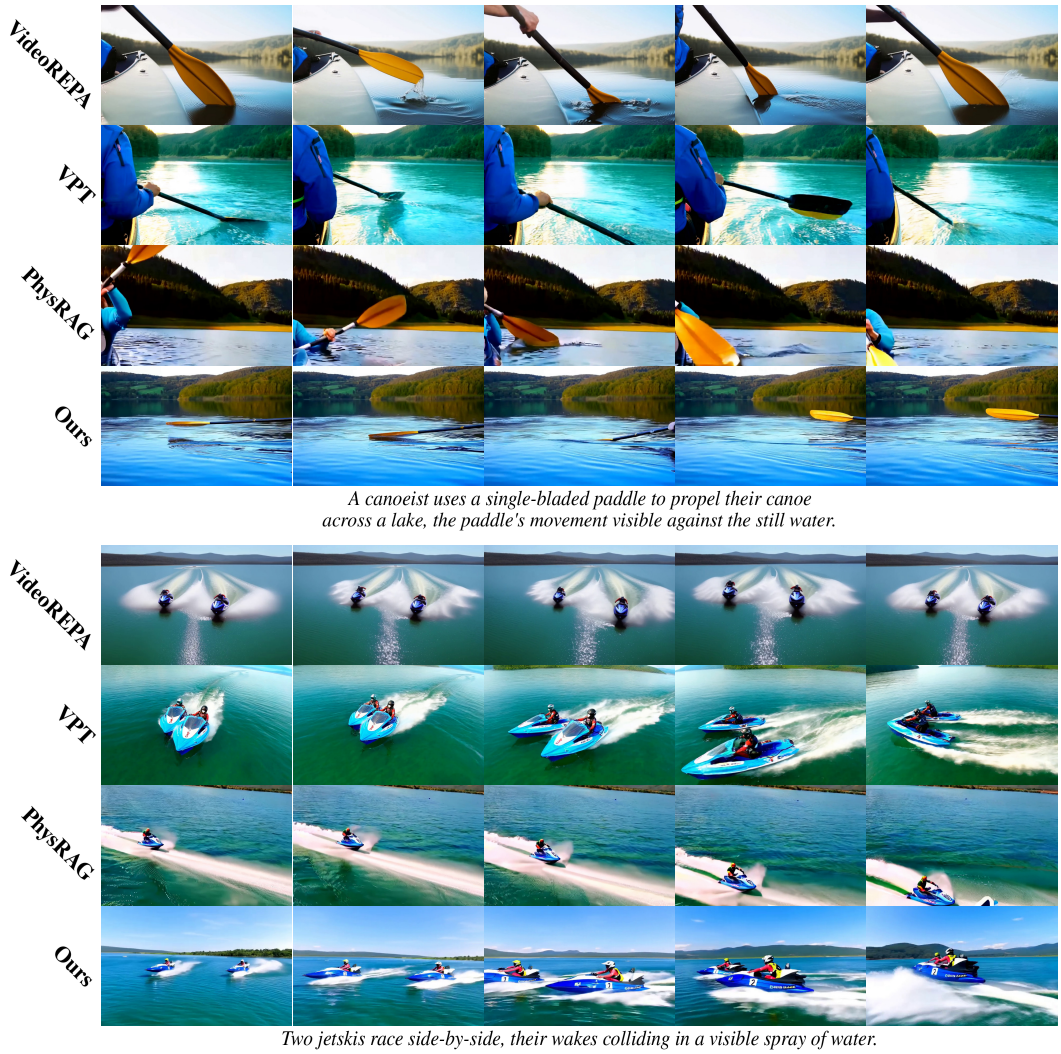

Figure 8: **Qualitative results on text-to-video generation.** The upper case shows a canoeist paddling across a lake, and the lower case shows two jetskis racing side by side. Within each case, rows show VideoREPA, VPT, PhysRAG and Ours from top to bottom. Each row contains five frames sampled at evenly spaced relative positions across the video, ordered from left to right.

Table 10: Component ablations at four-decimal precision (0–1, higher is better). Column abbreviations match Table 5. Colors indicate the top three scores in each column.

Method	Aesthetic Quality (AQ)	Background Consistency (BC)	Imaging Quality (IQ)	Overall Consistency (OC)	Subject Consistency (SC)	Quality Score (QS)
w/o time-step	0.5147	0.9236	0.6369	0.5850	0.9120	0.7970
w/o self-attention	0.4892	0.9213	0.6043	0.5812	0.9000	0.7854
w/o cross-attention	0.4848	0.9144	0.6066	0.5821	0.9022	0.7846
w/o hidden-state MLP	0.4928	0.9238	0.6359	0.5817	0.9086	0.7928
w/o refiner	0.5129	0.9249	0.6204	0.5914	0.9126	0.7863
v input	0.5038	0.9238	0.6368	0.6021	0.9174	0.7980
w/o condition	0.5154	0.9274	0.6123	0.5812	0.9102	0.7979
Ours	**0.5260**	**0.9359**	**0.6425**	**0.6179**	**0.9260**	**0.8026**

precision. The assumptions and complete proofs underlying the theoretical results in Section 4 are provided in Appendices A, B and C.

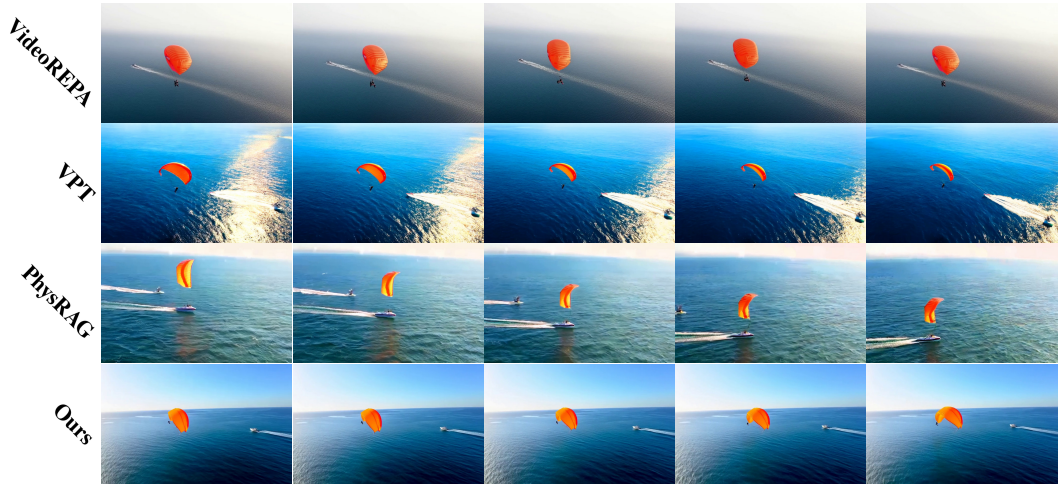

Figure 9: **Qualitative results on text-to-video generation.** The prompt describes a parasailer with an orange parachute being pulled across the ocean by a speedboat. Rows show VideoREPA, VPT, PhysRAG and Ours from top to bottom. Each row contains five frames sampled at evenly spaced relative positions across the video, ordered from left to right.

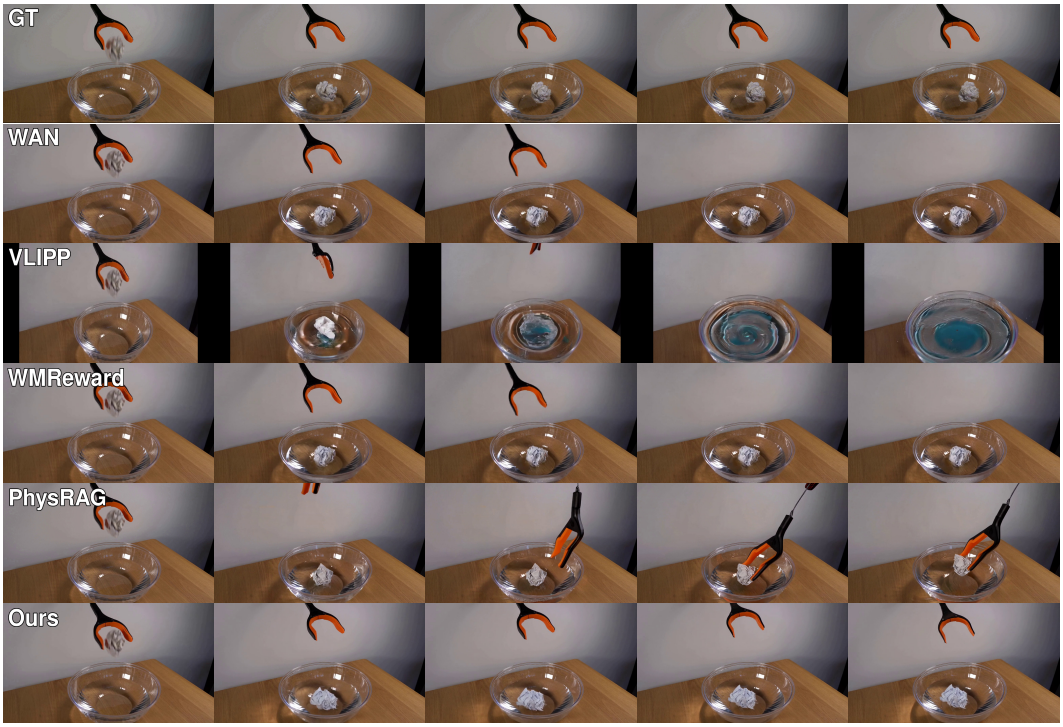

Figure 10: **Additional image-to-video comparison on Physics-IQ Verified: paper falling into water.** A grabber releases crumpled paper into a bowl of water. Rows show GT, WAN, VLIPP, WMReward, PhysRAG and CORRFLOW (Ours). Each row contains five video frames in temporal order.

Figure 11: **Additional image-to-video comparisons on Physics-IQ Verified: motion and reflections.** The upper case shows a suspended tennis ball being lowered in front of a mirror, and the lower case shows a tennis ball rolling toward a kettlebell beside a mirror. Within each case, rows show GT, WAN, VLIPP, WMReward, PhysRAG and CORRFLOW (Ours). Each row contains five video frames in temporal order.

Figure 12: **Additional image-to-video comparisons on Physics-IQ Verified: ball motion.** The upper case shows a bouncing tennis ball near a ramp, and the lower case shows a blue ball rolling from a table onto glass. Within each case, rows show GT, WAN, VLIPP, WMReward, PhysRAG and CORRFLOW (Ours). Each row contains five video frames in temporal order. Red boxes highlight visible balls and adjacent reflections in the four baselines after the initial frame.